Subject: Manufacturing Processes (23A0340IT)

UNIT - 1 : CASTING

Steps in casting, patterns & allowances, molding, cores, gating & risers, melting furnaces, solidification, defects, centrifugal / die / investment / shell casting

1. Introduction & Steps in Making a Casting

Casting is a manufacturing process in which molten metal is poured into a mould cavity having the shape of the desired part and is allowed to solidify. The solidified part is called a casting. It is one of the oldest shaping processes (about 6000 years old).

Casting = Melting + Pouring + Solidification + Removal + Finishing.

Steps involved in making a casting

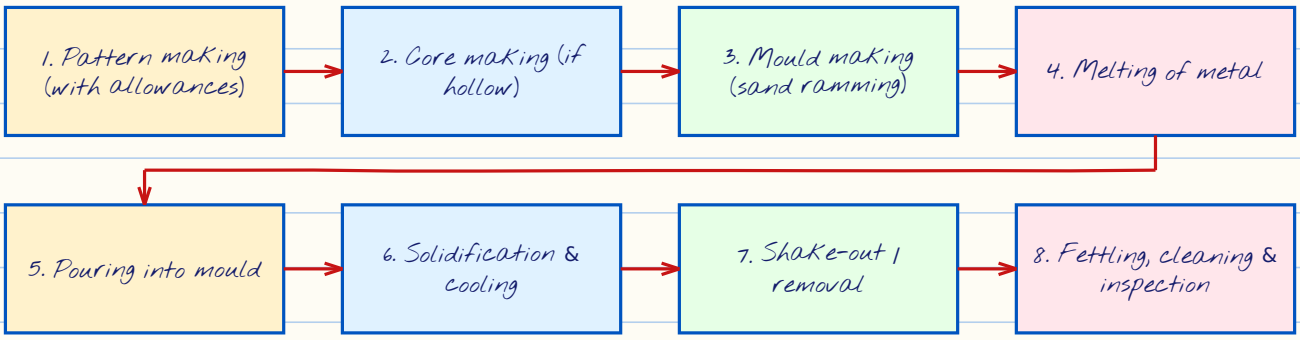


Fig 1.1 Sequence of operations in sand casting

- Pattern: replica of the part, made slightly larger (allowances) - used to make the cavity in sand.
- Mould: prepared by ramming moulding sand around the pattern in two halves - cope (upper) and drag (lower) - then pattern is withdrawn.
- Core: separate sand body placed in the cavity to form internal holes / hollow portions.
- Gating system: sprue, runner, gates and riser cutlplaced in the mould for metal flow & feeding.
- Fettling: removal of gates, risers, flash, sand; then grinding, heat-treatment & inspection.

Advantages of casting

- Very complex shapes (internal cavities, undercuts) can be made in a single piece.
- Almost any size - from a few grams (dental parts) to 100+ tonnes (ship propellers, machine beds).
- Practically any metal can be cast (ferrous & non-ferrous) - even those difficult to forge/machine.
- Low tooling cost for small quantities - economical for both one-off and mass production.
- Near-net shape: less machining & material wastage; scrap can be re-melted.
- Good vibration damping (cast iron) and isotropic properties; strong parts like engine blocks possible.

Limitations

- Dimensional accuracy & surface finish of sand casting are poor; machining often needed.
- Defects like porosity, blow holes, shrinkage are common; labour-intensive; mechanical properties are lower than wrought products.
- High-temperature metal handling - safety & pollution (fumes, silica dust).

Applications

Component	Examples
Automobile	Engine block, cylinder head, piston, manifolds, brake drums, gear-box housing
Machine tools	Lathe bed, machine bases, frames, gear blanks, pulleys, flywheels
Power & Heavy	Turbine casings, pump impellers, valves, boiler fittings
General	Pipes & fittings, manhole covers, sanitary ware, statues, art pieces, jewellery

2. Patterns & Pattern Making

A pattern is a model/replica of the object to be cast, used to form the cavity in the mould. It is normally made larger than the final casting to account for metal shrinkage and machining.

Functions of a pattern

- Forms the impression (cavity) of the casting in the mould.
- Carries core prints that locate cores; helps in establishing the parting line.
- Pattern also provides runners, gates & risers (in gated / match-plate type) and reduces defects.
- Helps in minimising overall cost - many moulds can be made from one pattern.

Types of patterns

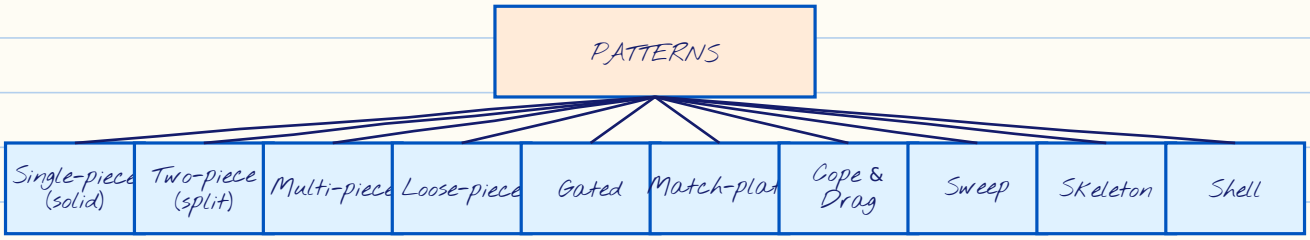


Fig 2.1 Classification of patterns

Type	Description & use
Single-piece / solid	One-piece, simplest, shape easy; parting line is flat. Used for simple shapes, low volume (e.g. flat plate, bush blank).
Split (two-piece)	Split at the parting line into 2 halves aligned by dowel pins. For moderately complex shapes - spindles, cylinders, pipes.
Multi-piece	Three or more parts (cope, cheek, drag) for very complex castings.
Loose-piece	Has projections/loose parts which are withdrawn separately after moulding - for shapes having undercuts.
Gated	Several patterns connected by gates/runners; mass production of small castings - saves moulding time.
Match-plate	Cope & drag patterns mounted on opposite sides of a metal plate, with gates; used with moulding machines; high accuracy & large quantity.
Cope & drag	Separate patterns for cope and drag mounted on separate plates; for heavy, big castings.
Sweep	A thin board rotated about a spindle to form large symmetrical moulds (bells, kettles) - saves pattern cost.
Skeleton	Wooden frame/skeleton (ribs); sand is filled & struck off - very large, few-off castings.
Shell (shell mould)	Metallic pattern, heated, for shell moulding process.

Materials used for patterns

Material	Merits	Demerits / Use
Wood (teak, pine, deodar, mahogany)	Cheap, light, easy to shape & repair	Absorbs moisture, wears quickly; for 1-100 castings
Metals (Al, brass, cast iron, steel)	Durable, accurate, smooth, high strength	Costly, heavy; mass production (Al most common)
Plastics (epoxy, polyurethane)	Light, no moisture, smooth finish	Less strength; medium run
Wax	Melts out easily, fine detail	Used in investment casting
Plaster of Paris	Intricate shapes, easy to cast	Weak, brittle; small castings
Rubber / Polystyrene	Flexible; expanded polystyrene vaporises	Used in full-mould / lost-foam

Pattern allowances

Dimensions of a pattern are changed from the casting dimensions by the following allowances:

Pattern Allowances (very important - 5 M)

- Shrinkage (contraction) allowance: metal shrinks on cooling from solidification temperature to room temp, so pattern is made larger. Positive allowance. Use a shrink rule (longer than a standard scale). Typical: grey C.I. 1 % (10 mm/m); steel 2 % (20 mm/m); aluminium 1.5 %; brass 1.6 %.
- Machining (finish) allowance: extra metal on surfaces to be machined (about 2-6 mm depending on size, metal & finish required).
- Draft (taper) allowance: slight taper (1-3 degree, or 10-20 mm/m on external surfaces; more on internal) on vertical faces so that the pattern can be withdrawn without damaging the mould.
- Shake (rapping) allowance: pattern is rapped to loosen it in the mould, which enlarges the cavity - so pattern is made slightly smaller (negative allowance, ~0.5-1 mm).
- Distortion allowance: for shapes like U, V, thin long casting which distort due to uneven cooling - pattern is given an opposite curvature.
- Core-print allowance: extra projection on pattern to provide seat for the core.

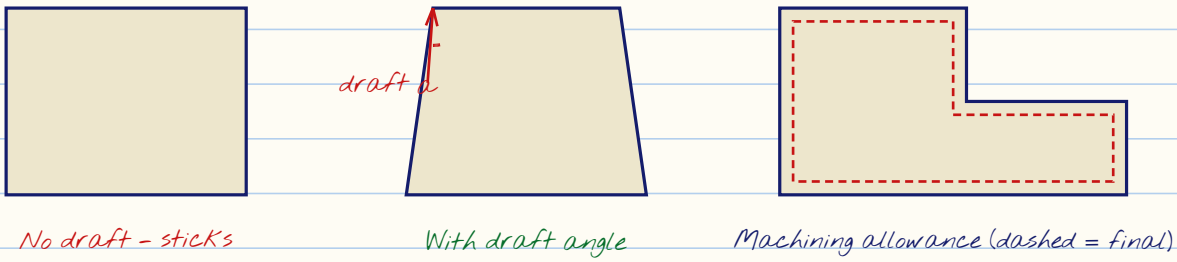


Fig 2.2 Draft & machining allowance

Pattern construction & colour code

- Wood patterns are built from well-seasoned wood joined by glue, screws, dowels; edges rounded with fillets (to avoid sharp corners, tearing & hot spots); surface varnished/shellac.
- Large patterns are built-up or skeleton; thick sections are hollowed to cut weight.
- Colour code:

Colour	Meaning
Red / Orange	Surfaces to be machined
Black	Un-machined (cast) surfaces
Yellow	Core prints / seats for loose pieces
Red stripes on yellow	Seats for loose pieces (stop-offs)
Black stripes on yellow	Stop-offs (spaces to be filled)

3. Molding (Moulding)

Moulding is the process of making a mould cavity by packing moulding sand around the pattern in a moulding box (flask) and withdrawing the pattern.

Moulding sand - composition & properties

- Composition: silica sand (SiO_2) 75-85 %, clay binder (bentonite / fire clay 10-20 %), moisture (2-8 %) and additives (coal dust, wood flour, dextrin, molasses).
- Types: green sand (moist, most common), dry sand (baked), loam, facing sand, backing sand, parting sand (dry silica to avoid sticking), core sand.

Properties of moulding sand

- Permeability: allows gases & steam to escape. - Refractoriness: withstand high temp without fusing.
- Cohesiveness: sand grains stick together. - Adhesiveness: sticks to the box.

- Flowability: fills pattern details. - Collapsibility: yields as casting shrinks (avoids hot tears).
- Green strength / Dry strength, good thermal stability, chemical inertness, reusability.

Steps in making a green-sand mould

- Place pattern (drag half) on moulding board in the drag box; sprinkle parting sand.
- Fill facing sand then backing sand; ram it uniformly; strike off excess; turn the drag over.
- Place cope half of pattern, sprue pin & riser pin; fill, ram, vent with vent wire (for gases).
- Remove sprue/riser pins, cut pouring basin; separate cope & drag; draw out the pattern gently (using draw spikes, rapping).
- Cut runner & gates; place cores; close the mould, clamp / weigh down the cope - ready for pouring.

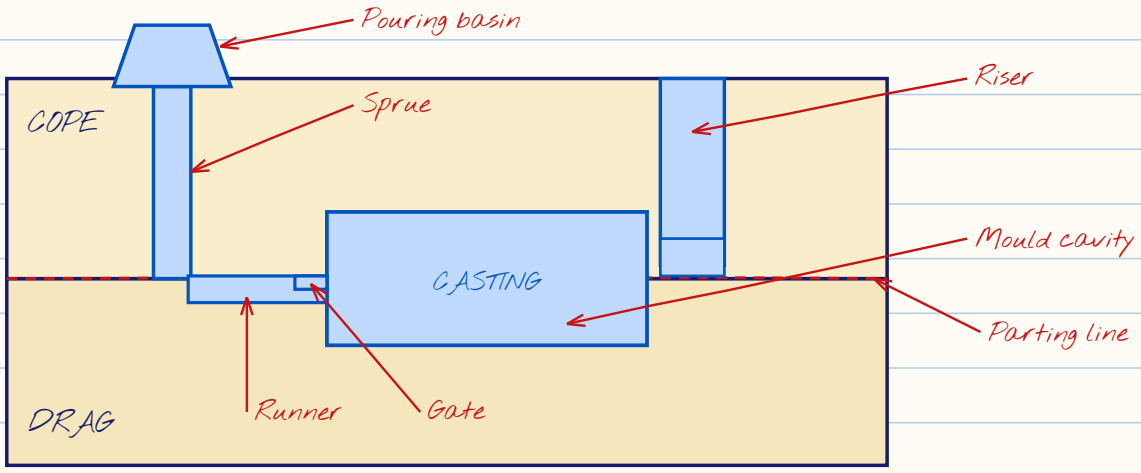


Fig 3.1 Sand mould with gating system

Types of moulding / moulds

- Bench / floor / pit moulding: small jobs on bench; medium on floor; very large castings in pits (pit itself acts as drag).
- Machine moulding: squeeze, jolt, jolt-squeeze, sand-slinger - for mass production, uniform ramming.

- Green sand mould: damp, cheapest, good collapsibility but gives lower strength & dimensional accuracy.
- Dry sand mould: baked at 150-300 degree C - stronger, more accurate, for large castings; costly, slow.
- Loam, CO_2, shell & no-bake moulds: loam for huge symmetrical parts; CO_2 process hardens sand (sodium silicate) quickly; no-bake uses resin binders.

4. Cores & Core Making

A core is a pre-formed body of baked sand placed in the mould cavity to produce internal cavities / holes / undercuts in the casting. Core sand contains silica sand + binders (linseed oil, core oil, cereal, resin, sodium silicate) - baked at 200-250 degree C to give high strength.

Requirements: adequate strength, permeability, refractoriness, collapsibility, smooth surface, low gas emission.

Types of cores

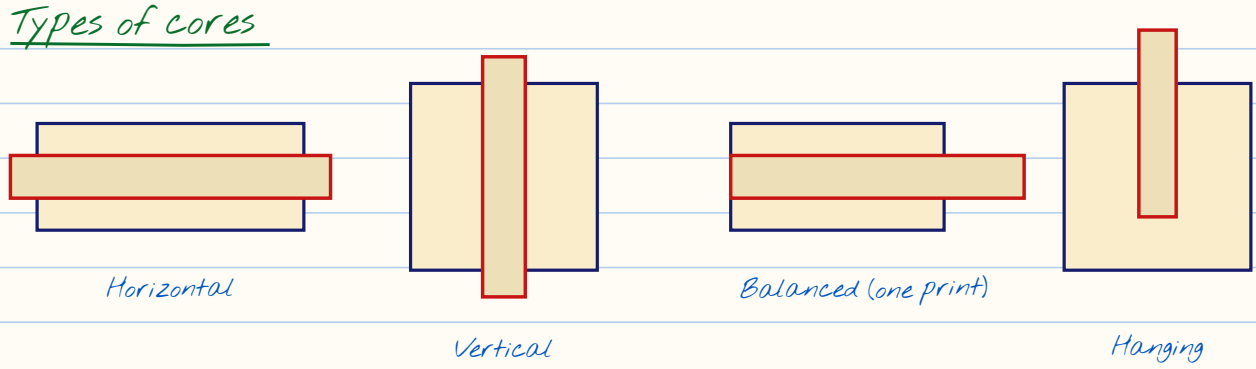


Fig 4.1 Some types of cores (red = core)

Core	Remarks
Horizontal	Placed horizontally; supported at both ends by core prints in the parting plane.
Vertical	Axis vertical; core prints on cope and drag - for holes in bushes, gear-blanks.
Balanced	Supported at only one end; its weight is balanced by a longer print.
Hanging (cover)	Hangs from cope; no support in drag. Cover core lies over the mould.
Drop / Ram-up	Drop core is placed from cope, below parting line. Ram-up core is rammed in with the pattern.
Kiss core	Core held in place by pressure between cope and drag, no core print.
Wing / Green-sand	Wing core for projections; green sand core formed by the pattern itself.

- Core prints: extra projections on pattern which form seats in mould to support the core.
- Chaplets: small metal supports (same material as casting) fixed between mould & core to prevent core sagging/floating; they fuse into the casting.
- Core making: sand mixed with binder, shaped in a core box by hand/ blowing machines (core blower / shooter), baked in core oven, finished & coated (blacking/ graphite), vented using wax wire.

5. Gating System, Risers & Casting Design

Gating system - principles

The gating system is the network of channels (pouring basin, sprue, runner, gates) that leads molten metal into the cavity. A good gating system should:

- Fill the cavity quickly & smoothly without turbulence, thereby avoiding oxidation, gas pick-up, erosion of sand (cuts & washes).
- Establish directional solidification (thin section/ far-end first, riser last).
- Trap slag, dross & inclusions (using skim bob, strainer core, choke, slag traps).
- Avoid air aspiration - the sprue should be tapered and have a well at the bottom.
- Maintain minimum metal consumption (high casting yield).

Elements of gating system

- Pouring cup / basin: receives metal from ladle, reduces turbulence & helps in skimming slag.
- Sprue: vertical tapered channel; velocity $v = \sqrt{2gh}$; continuity $A1\,v1 = A2\,v2$ so $A2/A1 = \sqrt{(h1/h2)}$ (tapered to avoid aspiration).
- Sprue well: enlarged base to absorb energy of falling metal.
- Runner: horizontal channel distributing metal to gates. - Gates (ingates): connect runner with cavity; control rate of filling, shape for easy removal.
- Riser: reservoir of molten metal feeding shrinkage.

Types of gating systems

- Top gate: metal enters from the top - promotes directional solidification but causes splashing/erosion (for short, simple castings).
- Bottom gate: smooth, quiet filling, little turbulence; but unfavourable temperature gradient.

- Parting-line (side) gate: most common compromise - at the parting plane; Step gate: multiple gates at different heights for tall castings.
- Pressurised vs un-pressurised: pressurised (gate area < runner area; ratio sprue:runner:gate = 1 : 0.75 : 0.5) - fast filling, for ferrous; un-pressurised (1 : 3 : 3 or 1:2:1) - less turbulence for Al/Mg alloys.
- Filling time $t = W / (\rho A_c v)$. Choke area $A_c = W / (\rho \cdot C \sqrt{2 g H})$.

Risers

A riser (feeder / feed head) is a reservoir of molten metal connected to the casting to compensate for liquid and solidification shrinkage. It must solidify after the casting. Functions: feeds shrinkage, lets out air/gases, indicates the mould is full, helps directional solidification.

- Types: top riser, side riser, blind riser (inside cope; has a atmospheric core - for better feeding), open / closed riser. Shape: cylindrical is best (min. surface area for volume - slow cooling).
- Chvorinov's rule: Solidification time $\{r|t = B (V/A)^2\}$ where V = volume, A = cooling surface area, B = mould constant. Riser must have larger V/A than the casting: $(V/A)_{riser} >= 1.2 (V/A)_{casting}$.
- Caine's method: freezing ratio $X = (A/V)_{casting} / (A/V)_{riser} = a / (Y - b) + c$, where Y = riser volume / casting volume.
- Chills: metallic inserts (internal / external) placed in the mould to accelerate cooling at thick sections and promote directional solidification; Padding: taper added to thin sections to help feed. Insulating / exothermic sleeves keep riser hot longer.

Casting design considerations

- Provide uniform section thickness; avoid abrupt changes (use gradual tapering, 1:4 ratio) - prevents hot spots & shrinkage cavities.
- Use fillets & rounded corners (no sharp edges - cracks & hot tears); avoid large flat surfaces (warpage - add ribs).
- Sections should join at proper angles; avoid intersection of many ribs at one point (hot spot) - stagger them.
- Choose parting line to be flat, simple & at the largest cross-section; provide draft.
- Design to minimise cores, avoid undercuts; provide machining allowance &

datum faces for machining.

- Locate thick parts near risers, to ensure directional solidification; avoid isolated heavy bosses.

6. Melting Methods & Furnaces

Furnace selection depends on type of metal, quantity, melting temperature, quality, cost & fuel. Common melting furnaces:

Furnace	Used for	Features
Cupola	Grey cast iron	Vertical shaft, continuous melting, cheap, coke fuel
Crucible (pit / tilting)	Non-ferrous (Al, Cu alloys), small	Oil / gas / coke fired; small batches; easy alloying
Electric arc	Steel, alloy steel, C.I.	Fast, high temp (up to 1800 C), good quality control, big capacity
Induction (coreless / channel)	Steel, C.I., non-ferrous	Clean, stirring by eddy currents, accurate composition
Reverberatory (air)	Malleable iron, Al scrap, large lots	Flame deflected from roof; large capacity, low cost
Pot / open-hearth	Steel (older)	Regenerative heating; slow but large

Cupola furnace

A vertical cylindrical furnace of steel shell lined with refractory (fire brick). Charge (pig iron + scrap + coke + limestone flux) is fed from the charging door at top in alternate layers; air blast through tuyeres near bottom burns coke and melts the iron which collects in the well and is tapped from the spout; slag is removed through the slag hole.

- Fuel : metallurgical coke (8–16 % of metal); flux limestone ($CaCO_3$) 2–3 % removes ash as slag.
- Capacity expressed in tonnes/hour of melt; blast rate ~ 30 m^3/min per m^2 of cross-section.
- Advantages: simple, continuous, economical, high melting rate. Limits: little control of composition (picks up S, loses Si/Mn), only cast iron.

Other furnaces

- Crucible: metal in clay-graphite / SiC pot heated externally; no contact with

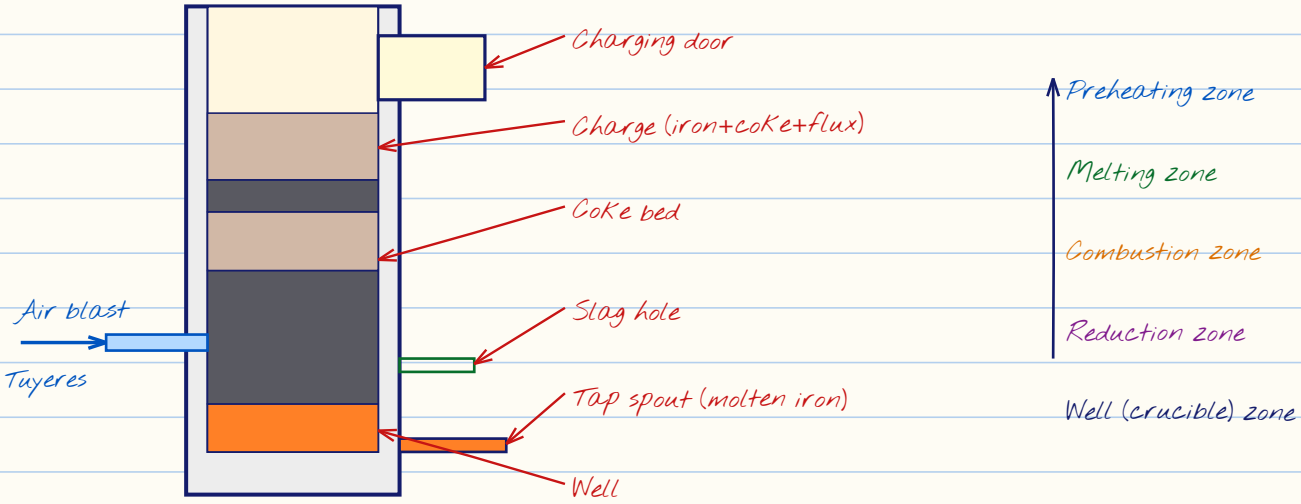


Fig 6.1 Cupola furnace

fuel => clean melt; tilting type pours directly.

- Electric arc (direct): three graphite electrodes strike arc with charge; temp up to 3000 C at arc; basic lining for steel (removes P,S).
- Induction: AC in a coil induces eddy currents in the charge placed in refractory crucible; stirring ensures uniform composition; low-frequency (channel) & high-frequency (coreless).

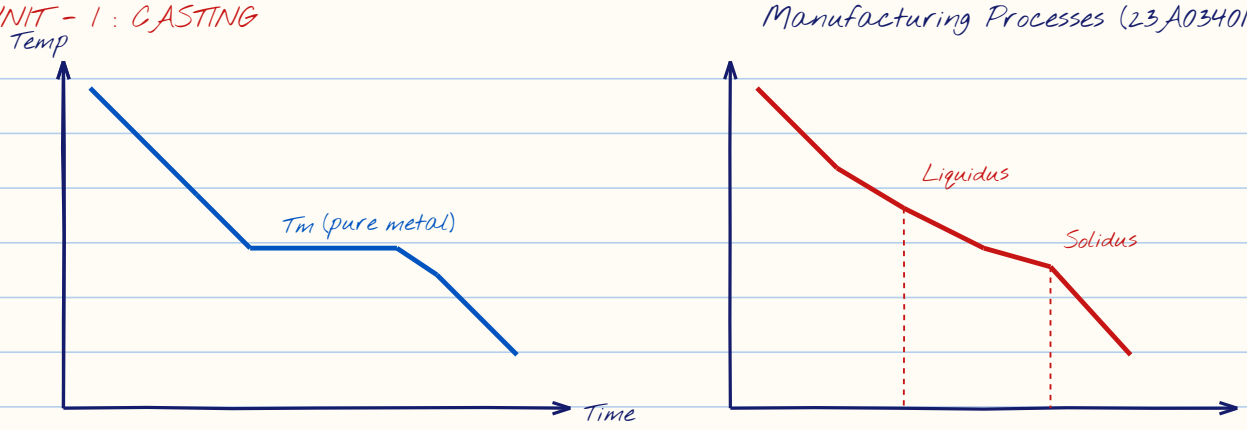
7. Solidification of Castings

- Pure metal: solidifies at a single temperature (constant); skin forms first at mould wall; columnar growth towards centre; shrinks and forms a pipe.
- Alloy: solidifies over a freezing range (liquidus - solidus) - mushy zone with dendrites; larger range => more porosity; skin may not form.
- Nucleation & growth: grain structure - chill zone (fine equi-axed at wall), columnar zone, central equi-axed zone. Fine grains give better strength: use inoculants / grain refiners (Ti, B in Al).
- Shrinkage: three stages - liquid contraction, solidification contraction (cavities), solid contraction (pattern allowance).
- Directional solidification: far end -> riser; progressive, using chills, risers, padding.

Chvorinov's rule (remember!)

Solidification time $t_s = B (V / A)^2$ (V = volume of casting, A = surface area cooling, B = mould constant)

- Riser should freeze later => riser V/A larger than casting V/A.
- A sphere has the highest V/A => longest time; thin plate shortest.



Pure metal - cooling curve

Alloy - freezing range

Fig 7.1 Cooling curves

8. Casting Defects - Causes & Remedies

Defect	Causes	Remedies
Blow holes / gas porosity	Trapped gases (moisture, high pouring temp, low permeability, poor venting)	Dry mould, increase permeability, proper venting, degas metal, lower pouring temp
Shrinkage cavity	Insufficient feeding, poor riser, non-uniform section	Proper riser size/location, chills, directional solidification, uniform section
Misrun	Metal freezes before filling cavity - low pouring temp, thin section, slow pouring	Raise pouring temp, increase fluidity & section thickness, improve gating
Cold shut	Two streams of metal meet but do not fuse (low temp / slow pouring)	Higher pouring temp, faster pouring, change gate position
Hot tear (crack)	Restricted contraction in a hot, weak casting; sharp corners	Collapsible sand/core, fillets, proper cooling, uniform sections
Cuts & washes	Erosion of sand by high-velocity metal	Reduce velocity, use strong sand, proper gating (well)
Inclusions / sand holes / dross	Loose sand, slag, dross entering cavity	Clean mould, skim, strainer/filters, good gating, proper ramming
Scab / swell / drop	Sand surface layer lifts due to hot metal; weak mould wall; soft ramming	Proper ramming, good permeability, correct moisture, strong sand
Mould/core shift	Misalignment of cope-drag, core displaced	Proper alignment (pins), clamps, chaplets, secure core
Fins / flash	Metal leaking along parting line (cope not weighted)	Clamp/ weight properly, good fit of box halves

Warpage / distortion	Uneven cooling, large flat areas	Ribs, proper pattern curvature, uniform cooling
Metal penetration / fusion	Very high temp, coarse sand, poor sand strength	Fine sand, mould wash/ coating (blacking), lower temperature

9. Special Casting Processes

9.1 Centrifugal casting

Molten metal is poured into a rotating mould; centrifugal force throws metal against the mould wall and it solidifies under high pressure giving dense, impurity-free castings (impurities, being lighter, collect at the bore & are machined away).

- True centrifugal: axis of rotation = axis of casting; hollow cylinders (pipes, liners, bushes) without cores. Horizontal (de Lavaud) / vertical axis. Speed N: G-factor (rw^2/g) 60–80 (horizontal).
- Semi-centrifugal: solid-type castings (wheels, pulleys, gear blanks) with a central core/riser; mould spins about vertical axis.
- Centrifuging: several small cavities arranged around a central sprue; metal forced into them by rotation – small, intricate parts (jewellery).
- Pros: no cores, dense fine grain, good mechanical properties, no gating loss. Cons: only symmetrical shapes; inner diameter inaccurate; segregation of alloys. Uses: pipes, cylinder liners, bushings, rolls.

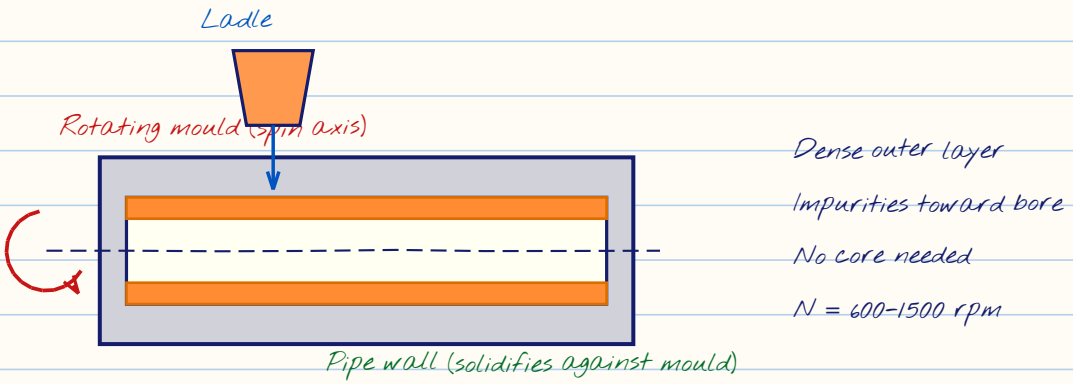


Fig 9.1 True centrifugal casting (horizontal)

9.2 Die casting (pressure die casting)

Molten metal is forced under high pressure (7-700 MPa) into a steel die (permanent mould) and held until it solidifies. Used mainly for low melting non-ferrous alloys - Zn, Al, Mg, Cu, Pb, Sn.

Hot-chamber	Cold-chamber
Injection cylinder (gooseneck) immersed in molten metal in the furnace; plunger forces metal into die.	Metal is ladled into a separate shot chamber, then plunger injects at high pressure.
For low-melting alloys (Zn, Sn, Pb); fast cycle (up to 500 shots/h); pressure 7-35 MPa.	For Al, Mg, Cu alloys (they attack the plunger/pot); pressures up to 150 MPa +; slower cycle.

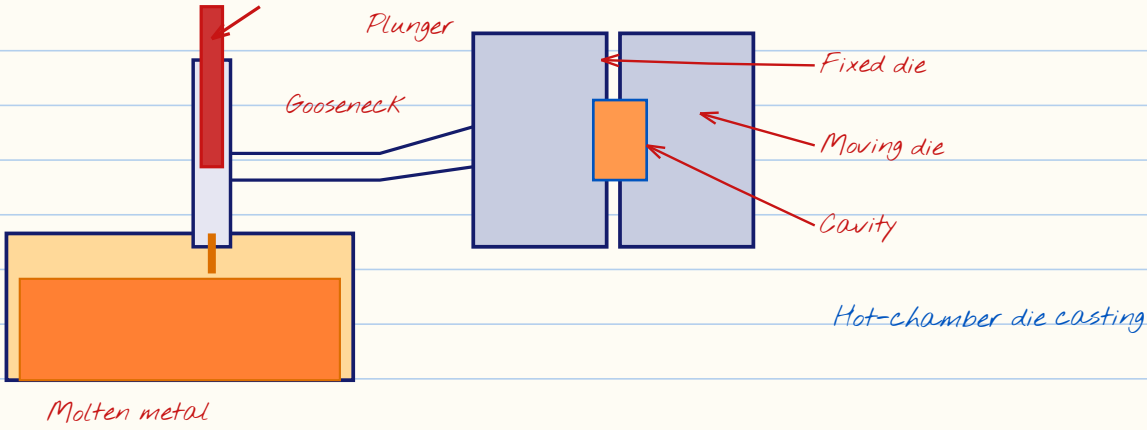


Fig 9.2 Hot-chamber die casting machine

- Advantages: high production rate, excellent dimensional accuracy & surface finish, thin walls, little machining, long die life.
- Limitations: high die/machine cost (economical only for large volume), size & metals limited, air entrapment porosity. Uses: carburettor, gear-box cases, toys, hardware, automobile & appliance parts.

9.3 Investment casting (lost-wax process)

Used for intricate, precise parts - turbine blades, jewellery, dental parts, surgical tools. Pattern is of wax which is melted out ("lost") from a ceramic shell.

- Merits: excellent accuracy (+-0.1 mm) & finish, extremely complex shapes, any alloy (even high-melting, superalloys), no parting line. Demerits: costly, slow, size limited (< 5-10 Kg usually), many steps.

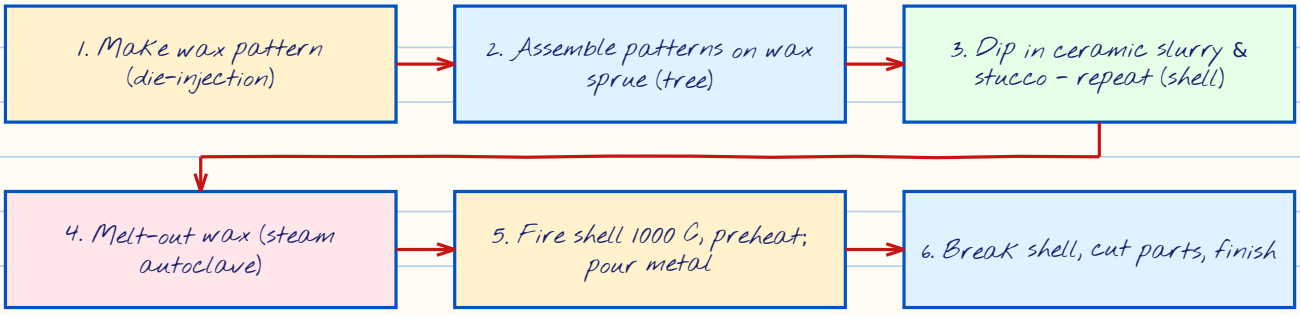


Fig 9.3 Steps in investment casting

9.4 Shell moulding (Croning process)

A thin shell (~5-10 mm) of sand + thermosetting resin (phenol-formaldehyde) is made over a heated metal pattern (200-250 C); shell halves are cured, joined and backed with sand/shot, then metal is poured.

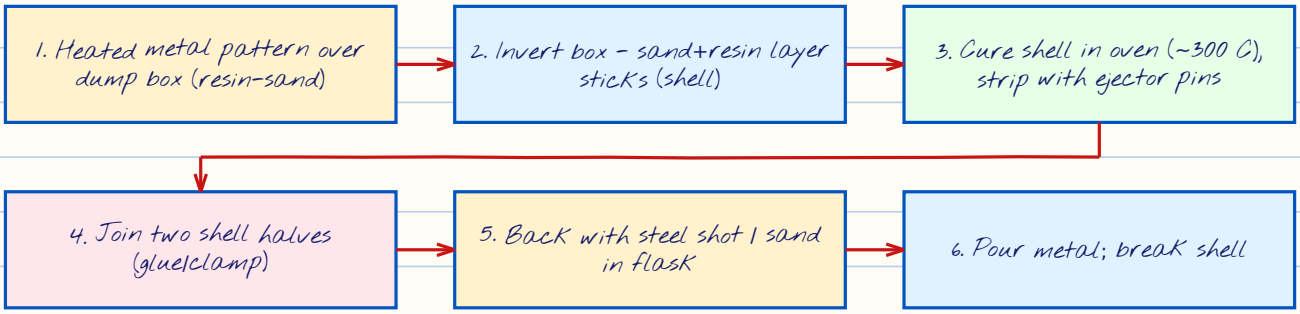


Fig 9.4 Shell moulding steps

- Advantages: good surface finish & accuracy (+-0.25 mm), thin sections, high production, low sand consumption, easy automation, no ramming skill. Limits: costly metal pattern & resin, size limited, fumes. Uses: gear housings, cylinder heads, connecting rods, camshafts, small automotive parts.

Important exam questions - Unit I

- Explain the steps involved in making a casting. List advantages & applications. (5 M)
- What is a pattern? Types of patterns & materials used. Explain allowances with examples. (10 M)
- Describe the principles of gating; design of sprue; types of gating systems. (10 M)
- Chvorinov rule & riser design; role of chills. (5 M)
- Explain the cupola furnace with a neat sketch. (10 M)
- Casting defects: causes & remedies (5-10 M).
- Centrifugal / die / investment / shell casting - principle, steps, merits & applications (10 M each).

Subject: Manufacturing Processes (23A0340IT)

UNIT - II : WELDING

Classification, welded joints, gas welding & cutting, arc welding (MMAW, SAW, TIG, MIG, ESW), resistance, friction, FSW, forge, explosive, thermit, plasma, laser, EBW, soldering & brazing, HAZ, defects

1. Introduction & Classification of Welding

Welding is a permanent joining process in which two or more parts are joined by coalescence (fusion of the interface) using heat, or pressure, or both, with or without filler metal. Examples: ship hulls, pressure vessels, pipelines, bridges, automobile bodies, rail tracks.

- Advantages: permanent, strong joint (efficiency up to 100 %), any shape/size, low cost, can be automated, portable equipment, can join dissimilar metals, less weight than riveting.
- Limitations: residual stresses & distortion, HAZ with changed properties, skilled labour, fumes & radiation hazards, joint not easily dismantled, defects need inspection.

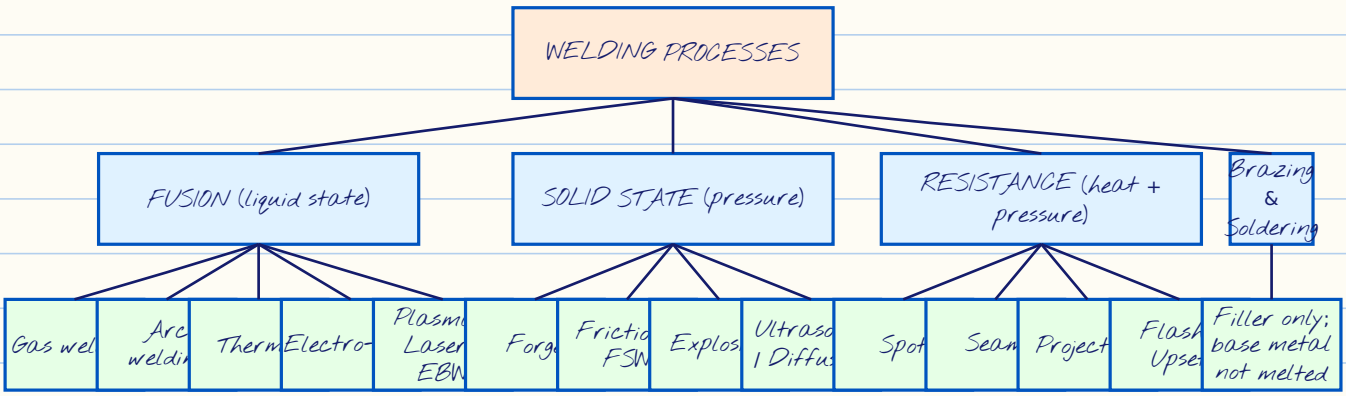


Fig 1.1 Classification of welding processes

Group	Principle	Examples
Fusion welding	Base metal melted (with/without filler), solidifies to form joint, no pressure	Gas, arc, thermit, ESW, plasma, laser, EBW
Pressure / solid-state	Metals heated to plastic state / cold & forced together by pressure; no melting	Forge, friction, explosive, ultrasonic
Resistance welding	Heat by I-squared-R at interface + pressure	Spot, seam, projection, butt (flash)

Brazing / soldering	Filler (m.p. below base metal) melts & flows by capillary action	Torch, furnace, dip, induction

2. Types of Welded Joints & Characteristics

Five basic joint configurations: butt, lap, corner, tee (T) and edge joints. Welds made on these joints are groove, fillet, plug/slot, spot, seam, and surfacing welds.

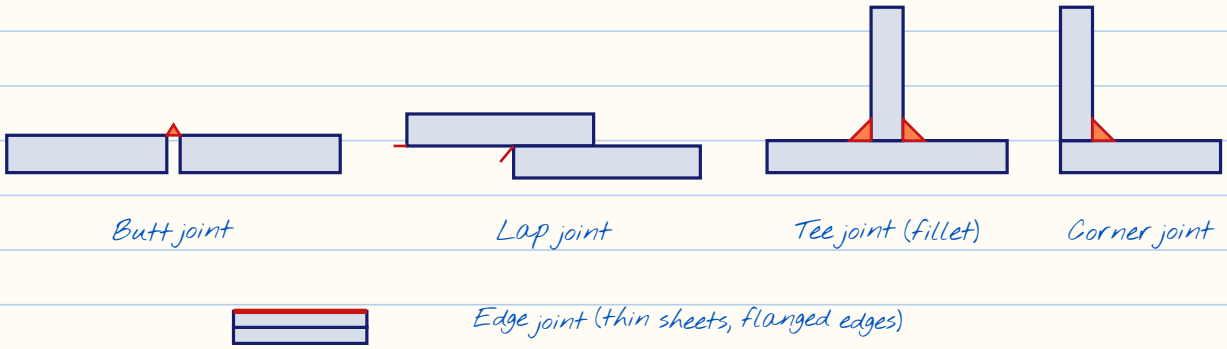


Fig 2.1 Basic weld joints (red = weld metal)

Joint	Characteristics / use
Butt	Two plates in same plane, edges butted. Strongest, best fatigue strength; needs edge preparation for thick plates (square, single/double V, U, J, bevel). Used in pipes, pressure vessels, ship plates.
Lap	Overlapping plates, fillet welds on one/both sides. Simple, no edge prep; strength 70–80 % of butt; eccentric load so bending; used for thin sheets, body panels.
Tee (T)	One plate perpendicular to another; fillet or groove weld; used for stiffeners, frames, beams.
Corner	Two plates meeting at right angle at edges – boxes, frames, tanks.
Edge	Plate edges aligned/parallel, weld along edges – sheet metal, light duty, no load carrying.

- Groove types: square ($t < 5$ mm), single-V (up to 20 mm, 60–90 degrees), double-V (thick plate, less distortion), U / J (very thick – less filler), bevel.
- Fillet weld: triangular section; leg size, throat = 0.707 x leg; strength = 0.707 x leg x length x allowable shear stress.

3. Gas Welding (Oxy-Acetylene)

Heat is produced by burning a fuel gas (acetylene) in oxygen. The flame melts the edges; a filler rod is added. Most common: oxy-acetylene (flame up to ~3200 C). Other fuels: hydrogen, LPG, propane, natural gas (lower temperature - for brazing, soldering).

- Equipment: oxygen & acetylene cylinders (O_2 - black / acetylene - maroon; acetylene dissolved in acetone, max 1.5 bar), pressure regulators, hoses (red-acetylene, black/blue-oxygen), blow-pipe (torch) with mixing chamber & nozzle, flash-back arrestors, goggles, spark lighter.
- Reaction: $C_2H_2 + O_2 \rightarrow 2CO + H_2$ (primary, inner cone, 3200 C); then $2CO + H_2 + 1.5\ O_2 \rightarrow 2CO_2 + H_2O$ (secondary, outer envelope).
- Systems: high-pressure (> 0.1 bar acetylene - cylinder) and low-pressure (injector type, from generator).
- Techniques: leftward (forward) for thin plates (< 5 mm); rightward (backward) for thicker plates (> 5 mm) - better heat use, less filler; vertical & overhead positions.
- Flux: borax for brass/ bronze, C.I., Al flux, to remove oxides.

Types of oxy-acetylene flames

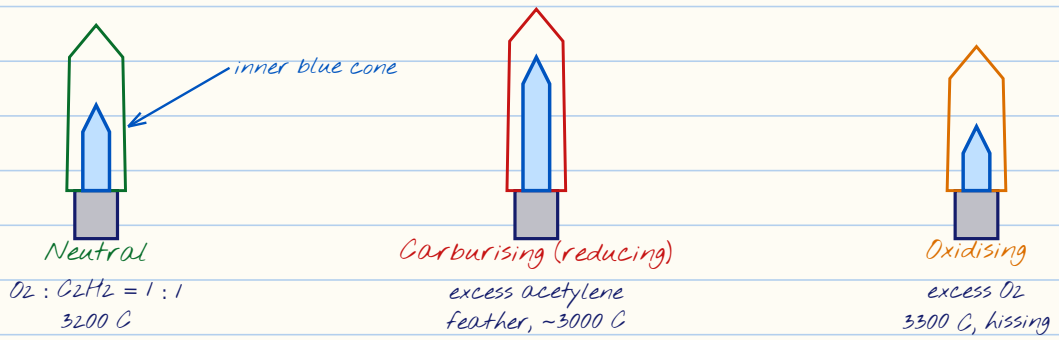


Fig 3.1 Oxy-acetylene flames

Flame	Appearance	Uses
Neutral (1:1)	Clear sharp inner cone, no feather; soft	Most welding: mild steel, C.I., stainless, Cu, Al - no effect on molten metal
Carburising / reducing (excess acetylene)	Three zones; whitish feather around inner cone	Hard surfacing (stellite), monel, nickel, high-carbon steel, Al; low-temp brazing; adds carbon - not for mild steel

Oxidising (excess O2)	Short, bluish, small cone; hissing sound	Brass, bronze, brazing, zinc-base alloys (forms protective oxide layer); harmful to steel (burning, porosity)

Oxy-acetylene gas cutting

Cutting of ferrous metals by rapid oxidation (burning). Steel is pre-heated to about 800-900 C (Kindlingignition temp.) by neutral flames; a jet of pure oxygen is then directed on the spot: Fe + O2 -> iron oxide (Fe3O4 + heat), which is blown away as slag, forming a kerf. Reactions are exothermic so cutting continues.

- Torch has pre-heating holes around a central oxygen cutting jet. Thickness: up to 300 mm+ steel.
- Works for steels and low-alloy steel; not for C.I., stainless steel, Al, Cu (oxide m.p. higher than metal, or conduction too high) - use plasma cutting / flux injection.
- Conditions: ignition temp < m.p. of metal; oxide m.p. < metal m.p.; low thermal conductivity.

4. Arc Welding - Basic Principles

An electric arc (sustained discharge through ionised gas between electrode & work) provides heat of about 6000-7000 C at the arc column; metal melts at 1500+ C. Arc voltage 20-40 V, current 50-1000 A. Electrode may be consumable (coated rod, wire) or non-consumable (tungsten, carbon).

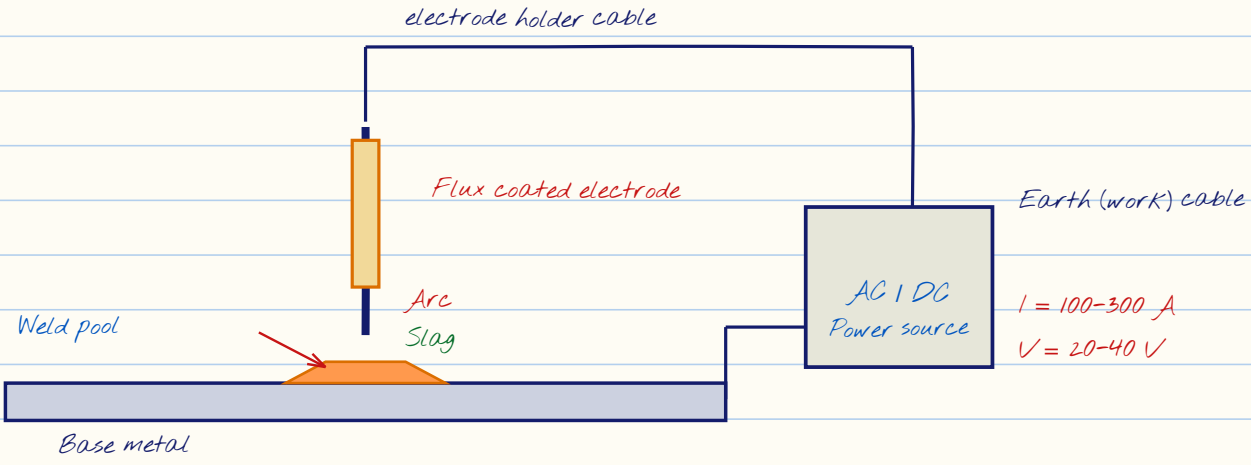


Fig 4.1 Arc welding circuit (SMAW)

- Polarity: DCSP (straight, electrode -ve): 2/3 heat at work - deep penetration, thick plates; DCRP (reverse, electrode +ve): 2/3 heat at electrode - shallow

penetration, thin sheets, Al/Mg (cathodic cleaning of oxide); AC - 50:50, avoids arc blow, for Al with HF.

- Arc blow: magnetic deflection of arc (DC) - reduce by AC, earth connection placement, short arc.
- Heat input: H = (V x I x 60 / S) x eta J/mm (S = travel speed mm/min, eta = 0.7-0.9).
- Power sources: welding transformer (AC, cheap, 20 KVA); rectifier (AC to DC); motor-generator set; inverter (compact, efficient). Duty cycle = (arc-on time / 10 min) x 100.

Power characteristics (V-I curves)

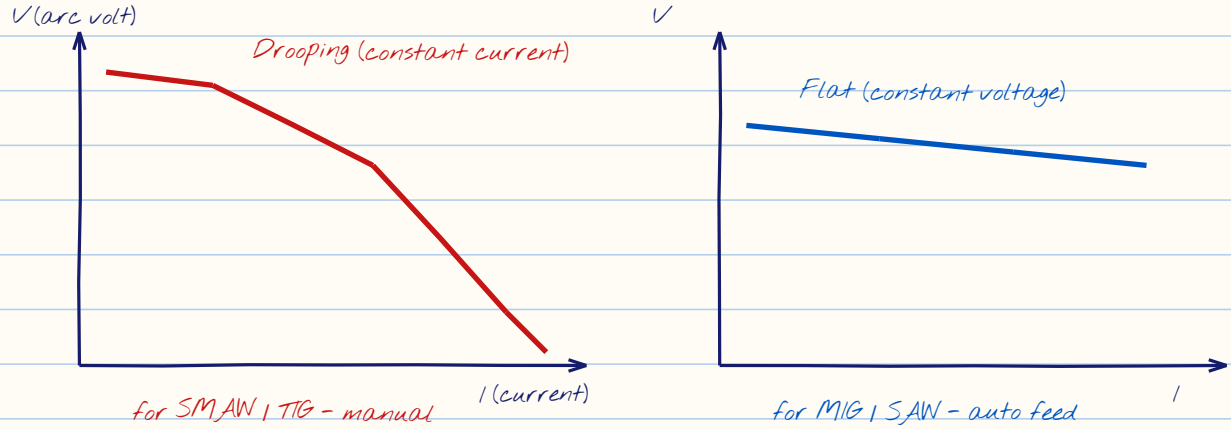


Fig 4.2 Static V-I characteristics

- Drooping (constant-current): large change in arc length (voltage) causes small change in current - manual welder can vary arc length without affecting weld; high OCV (60-80 V) for striking.
- Flat (constant-voltage): arc length self-regulates; current rises sharply if wire stick-out reduces; burn-off adjusts wire feed - used for semi-automatic MIG/CO2/SAW with constant wire feed.

5. Arc Welding Processes

5.1 Manual Metal Arc Welding (MMAW / SMAW / stick)

A flux-coated consumable electrode is held by hand; arc between electrode & work melts both; coating burns to give shielding gas & slag. Electrode specification e.g. E 6013: E = electrode; 60 = tensile strength x 1000 psi (60 Ksi); 1 = position (all); 3 = coating & current type.

Functions of electrode coating (flux)

- Produces shielding gas (CO_2, CO) - protects molten pool from O_2/N_2. - Forms slag that covers weld - slow cooling, protects from oxidation.
- Stabilises the arc (K, Na compounds); adds alloying elements (Mn, Cr) & deoxidisers (Fe-Mn, Fe-Si).
- Improves deposition rate (iron powder), controls bead shape. Types: cellulosic, rutile, basic (low-hydrogen), acid.

- Merits: simple, cheap, portable, all positions, wide range of metals. Demerits: low productivity, stub-end loss (about 50 mm), slag removal, needs skill, fumes.

5.2 Submerged Arc Welding (SAW)

A bare consumable wire is fed continuously into a blanket of granular flux deposited ahead of the arc; the arc burns submerged (invisible, no spatter). Unused flux is recovered by a vacuum; slag peels off.

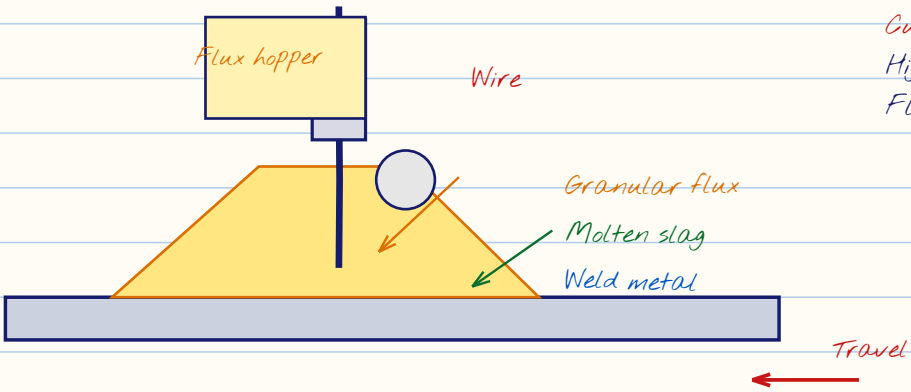
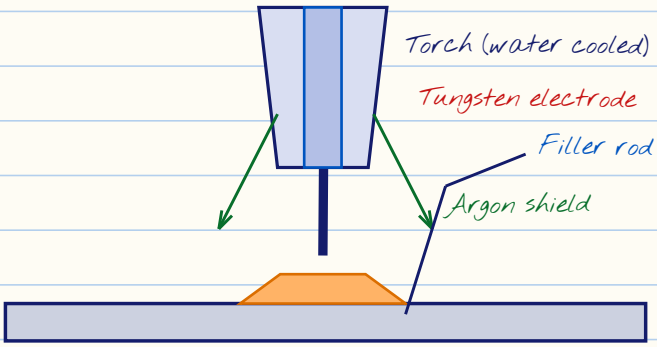


Fig 5.1 Submerged arc welding

- Merits: very high deposition (up to 5x SMAW), deep penetration, smooth uniform bead, no arc flash/spatter, high quality, automatic. Limits: only flat/horizontal, long straight seams; flux handling; thick plate. Uses: pressure vessels, ship panels, pipes, structural girders, tanks.

5.3 TIG welding (GTAW)

Tungsten Inert Gas: arc between a non-consumable tungsten electrode (2 % thoriated / ceriated) and work, shielded by inert gas (Ar / He). Filler rod added separately by hand (optional). DCSP for steel; AC + high-frequency for Al & Mg. Water/gas-cooled torch.

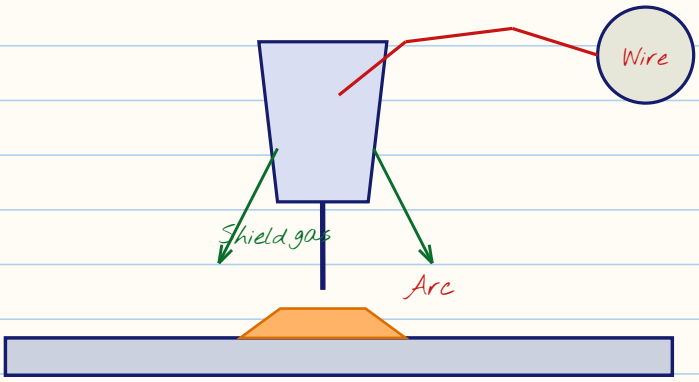


Clean welds, no slag / spatter
Thin sections, Al, Mg, Ti, stainless
Slow, costly, tungsten inclusion risk

Fig 5.2 TIG welding

5.4 MIG welding (GMAW)

Metal Inert Gas (MIG - Ar/He) or MAG (CO_2 / Ar-CO_2 mixes). A continuously fed consumable bare wire electrode with shielding gas; constant-voltage DCRP power. Semi-automatic gun; metal transfer modes: short-circuit (dip), globular, spray, pulsed.



Fast, high deposition, easy to automate
Steel, Al, Cu, stainless; all positions
Gas shield spoiled by wind; costly set-up

Fig 5.3 MIG / MAG welding

Feature	TIG	MIG
Electrode	Non-consumable tungsten	Consumable wire
Filler	Separate rod	Wire itself
Shield gas	Ar, He (inert)	Ar, He, CO_2, mixes
Quality / speed	Very high quality; slow	Good quality; fast
Use	Thin sheet, reactive & exotic metals	Production welding, structures, automobile

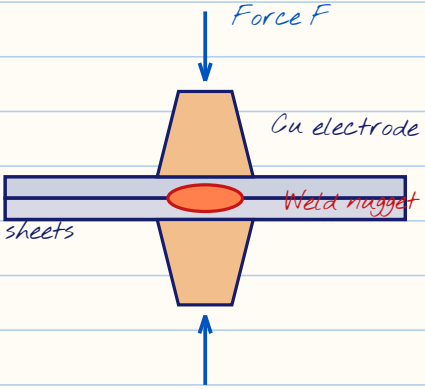
5.5 Electro-slag welding (ESW)

For thick plates (20 mm - 1 m) in vertical position, single pass. Plates are kept with a gap of 25-35 mm; water-cooled copper shoes (dams) both sides. Arc is struck initially under flux; once a slag pool forms, arc extinguishes and heat is produced by resistance heating of conductive molten slag ($I \times I \times R$). Consumable wire (or plate guide) is fed into the pool; weld metal solidifies upwards.

- Merits: single pass for very thick plates, no edge preparation, high deposition (15-20 Kg/h), low distortion. Limits: only vertical, coarse grain & low toughness in HAZ (normalising needed), cannot stop midway. Uses: heavy pressure vessels, ship hulls, heavy machinery frames, bridges.

6. Resistance Welding

Heat is produced by resistance to flow of current across the joint interface: $H = I^2 R t$ (joules). Pressure is applied by electrodes (Cu alloy). Low voltage (1-10 V), very high current (3000-100000 A), time 0.02-few s. No filler, no flux.



Spot welding of sheets
Squeeze - weld - hold - off
Seam: rollers instead of electrodes
(continuous overlapping spots, leak-tight)

Fig 6.1 Resistance spot welding

Type	Description	Use
Spot	Overlapping sheets clamped between two Cu electrodes; nugget forms at interface (3-5 sqrt t mm dia.)	Car bodies, sheet metal, appliances
Seam	Roller electrodes -> continuous seam of overlapping spots	Fuel tanks, drums, gas-tight vessels
Projection	Projections embossed on one part concentrate current; many spots at once	Nuts, bolts, studs on plates
Butt (upset)	Ends pressed together, current heats, upset forgings	Wire, rods, rings
Flash	Arcing (flashing) between ends then forging pressure; metal expelled	Rails, chains, tubes, large sections

- Merits: fast, automatic, no filler, low distortion, clean. Limits: high equipment cost, thickness limit, lap joints only (spot), skilled set-up.

7. Solid-State & Special Welding Processes

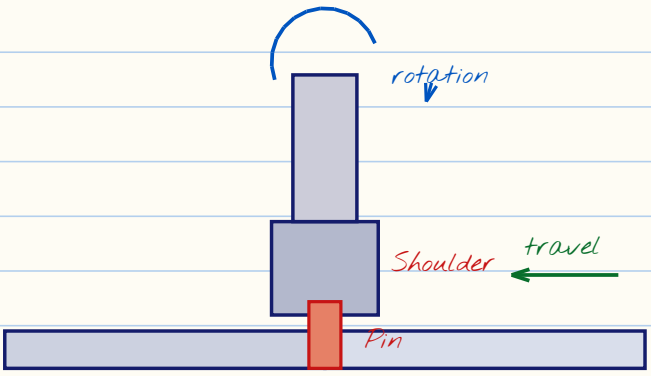
7.1 Friction welding

One part is rotated at high speed against a stationary part under axial force. Frictional heat raises interface to plastic state; rotation stopped and upset force applied -> forged joint, flash expelled. No filler, flux; no melting.

- Types: continuous-drive, inertia (flywheel), linear friction. Rotary - round bars/tubes.
- Merits: strong joint, dissimilar metals (steel-Al, Cu-steel), fast (seconds), no shielding, narrow HAZ. Uses: engine valves, drill bits, shafts, axles, pump parts. Limit: mainly round sections; costly machine.

7.2 Friction stir welding (FSW)

A rotating, non-consumable tool with shoulder and pin is plunged into the joint line of two butted plates and traversed. Friction & plastic deformation heat the metal (below melting); the pin stirs softened material from the advancing to retreating side forming a solid-state joint.



Al, Mg, Cu alloys; aerospace panels
No melting - no porosity, low distortion
Keyhole at end; heavy clamping needed

Fig 7.1 Friction stir welding

7.3 Forge welding

Oldest process: parts heated in a forge to white heat (about 1300 C), overlapped, and hammered / pressed to join (flux - sand/ borax to dissolve oxides). Used for wrought iron, low carbon steel; chains, tools; now replaced by modern methods.

7.4 Explosive welding

A controlled explosive charge drives a flyer plate at high velocity (~ 300-800 m/s) onto a base plate at a small angle (standoff); the oblique impact produces a metal jet that removes oxide films and creates a metallurgical (often wavy) bond - solid state, no melting.

- Joins dissimilar metals not weldable otherwise (Al-steel, Ti-steel, Cu-steel); cladding of large plates, transition joints; tubes to tube sheets. Limit: safety, noise, simple geometry.

7.5 Thermit welding

Exothermic reaction of aluminium powder + iron oxide (thermit): $8Al + 3Fe_3O_4 \rightarrow 9Fe + 4Al_2O_3$ + heat (2500-3000 C). Superheated molten iron is poured into a mould built around the joint; it fuses with the preheated ends. Slag (Al_2O_3) floats. Uses: rail joints, heavy section repairs (shafts, frames), crankshaft.

7.6 Plasma arc welding (PAW)

Constricted arc - gas (Ar) is forced through a small water-cooled copper nozzle producing a high energy-density plasma jet (up to 20000 C). Modes: transferred (arc to work) and non-transferred (arc inside torch). Keyhole mode for thick plates (up to 10 mm single pass).

- Merits: stable arc, narrow weld, deep penetration, high speed, less HAZ; also used for plasma cutting & spraying. Limits: expensive, noise, UV, torch life.

7.7 Laser beam welding (LBW)

A highly focused coherent beam (CO_2 or Nd:YAG / fibre laser, 1-100 KW) melts metal; Keyhole effect gives deep, narrow welds. No vacuum needed; shielding gas used. Heat input low -> small HAZ.

- Merits: precise, non-contact, high speed, automation (robots), welds thin / dissimilar metals, micro-welding. Limits: very high equipment cost, reflective metals (Al, Cu) difficult, joint fit-up must be exact, low efficiency. Uses: automotive tailor-welded blanks, electronics, medical, turbine parts.

7.8 Electron beam welding (EBW)

A beam of high-velocity electrons (30-150 KV) in a vacuum chamber (10^{-4} to 10^{-6} torr) strikes the joint; Kinetic energy converts to heat. Keyhole forms - very deep,

narrow weld (depth : width up to 30:1).

- Merits: highest quality, deep penetration (up to 300 mm), minimal distortion & HAZ, reactive metals (Ti, Zr, refractory), no flux/filler. Limits: vacuum chamber (size limit, time), costly, X-ray hazard. Uses: aerospace (jet engine), nuclear, gears, precision parts.

8. Soldering & Brazing

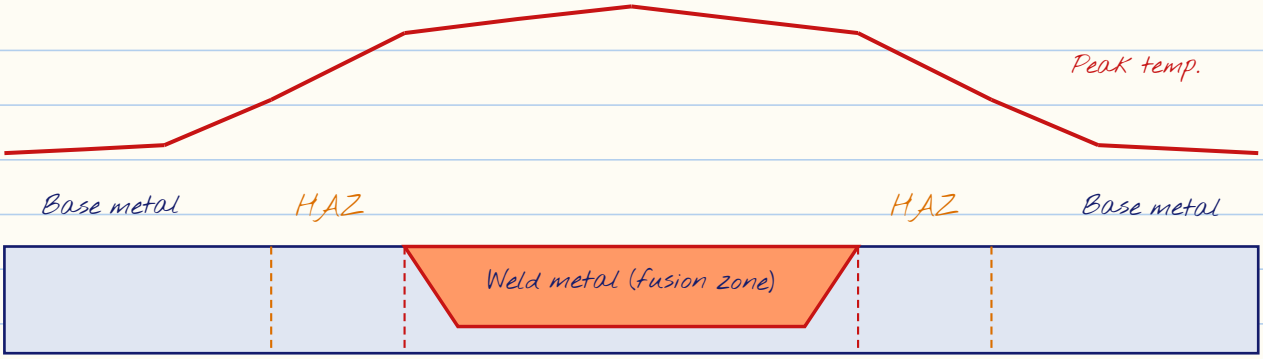
Feature	Soldering	Brazing
Filler temperature	Below 450 C (typically 180-300 C)	Above 450 C, below base-metal m.p. (600-1100 C)
Filler metal	Sn-Pb, Sn-Ag, Sn-Cu (lead-free) alloys	Brass / bronze (Cu-Zn), silver alloys, Al-Si, Cu-P
Flux	Rosin, zinc chloride, ammonium chloride	Borax, boric acid, fluorides
Joint strength	Low (30-50 MPa); electrical contact	Moderate-high (200-400 MPa); load carrying
Joint design	Lap, close fit	Lap/butt; clearance 0.025-0.25 mm for capillary action
Uses	Electronic PCBs, tin cans, radiators, pipes	Tools tips, bicycle frames, heat exchangers, pipes, carbide tips

- Heating methods: torch, furnace, dip, resistance, induction, infra-red. Base metal does NOT melt; filler spreads by capillary action; fluxes remove oxide; clean surfaces essential.
- Braze welding: filler is deposited like a fusion weld (no capillary) - repair of C.I.

9. Heat Affected Zone, Pre/Post Heating

The Heat Affected Zone (HAZ) is the portion of base metal which is not melted but whose microstructure & properties are altered by welding heat.

- Width of HAZ increases with heat input, plate thickness, preheat; decreases with high travel speed & low heat input (laser/EBW have narrowest HAZ).
- Problems: hardening (martensite in steels) -> cracks; grain coarsening -> poor toughness; softening in cold-worked or age-hardened alloys.



HAZ zones: coarse grain (near fusion line) -> fine grain (normalised) -> partially transformed -> unaffected

Fig 9.1 Weld zones and peak temperature

Pre-heating & Post-heating

- Pre-heating (150–300 C+): reduces cooling rate -> less martensite & residual stress, removes moisture, avoids cold (hydrogen) cracking; needed for high-carbon, alloy steels, thick sections, cast iron.
- Post-heating / PWHT: stress relieving (550–650 C for steel), tempering, normalising; reduces residual stresses, improves toughness, hardness & dimensional stability; hydrogen bake-out.
- Carbon equivalent $CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15$: higher CE (> 0.4) => weldability poor, preheat needed.

10. Welding Defects - Causes & Remedies

Defect	Causes	Remedies
Porosity / blow holes	Gas trapped (moisture, dirt, oil, wrong electrode, damp flux, long arc)	Clean/ dry, low-H electrodes, shorter arc, correct gas flow, slow cooling
Slag inclusion	Slag trapped between passes; poor cleaning; low current	Clean slag after each pass, proper manipulation, higher current
Lack of fusion	Low current/heat, high speed, wrong angle	Increase current, reduce speed, correct angle & joint design
Incomplete penetration	Narrow root gap, low current, thick root face	Wider gap, higher current, better joint preparation
Undercut	High current, long arc, fast travel	Reduce current/ arc length, steady travel, correct angle
Overlap	Low travel speed, too low current, wrong electrode angle	Increase speed & current

Cracks (hot / cold)	High residual stress, high CE, hydrogen, rapid cooling, S & P	Pre-heat, low-hydrogen electrodes, proper sequence, PWHT, ductile filler
Spatter	High current, long arc, arc blow, damp electrode	Reduce current/ arc, dry electrodes, anti-spatter spray
Distortion / warpage	Uneven heating & contraction	Jigs, tack welds, back-step, symmetrical welding, pre-set, stress relief
Burn-through	Excess heat, thin metal, big root gap	Lower current, backing strip, quicker travel
Crater / poor bead	Sudden arc break; wrong technique	Fill crater, run-off tabs

Important exam questions - Unit II

- Classify welding processes. Types of weld joints & their characteristics. (10 M)
- Oxy-acetylene welding: flames & their uses; gas cutting principle (10 M).
- Arc welding: principle, V-I characteristics, polarity; MMAW electrode coating (10 M).
- Explain SAW / TIG / MIG / ESW with neat sketch & applications (5-10 M each).
- Resistance welding types; friction, FSW, explosive, thermit, plasma, laser & EBW (5 M each).
- Brazing vs soldering; HAZ; pre & post heating; welding defects, causes & remedies (10 M).

Subject: Manufacturing Processes (23A0340IT)

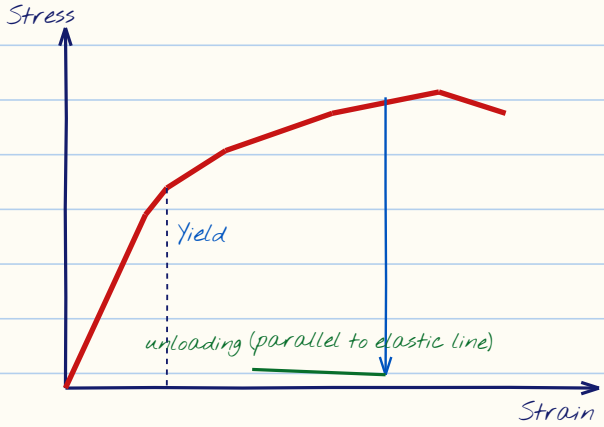
UNIT - III : BULK FORMING

Plastic deformation, recovery-recrystallization-grain growth, hot & cold working, strain hardening, annealing, forging, rolling, extrusion, wire & tube drawing

1. Plastic Deformation in Metals & Alloys

Bulk forming (metal working) changes the shape of a solid billet by plastic deformation - volume is conserved and the surface-area-to-volume ratio changes. Metal is stressed beyond its yield point.

- Elastic deformation: reversible, atoms displaced slightly; obeys Hooke's law ($\sigma = E \times \varepsilon$).
- Plastic deformation: permanent; occurs by slip (movement of dislocations along slip planes & directions) and twinning (mirror-image shear, mostly in HCP/BCC at low temp. & high strain rates).
- Real metals deform at much lower stress than theory because of dislocations; slip occurs on the close-packed planes (FCC - 12 slip systems -> most ductile; HCP - 3 -> less ductile; BCC - 48).
- Critical resolved shear stress: $\tau = \sigma \cos(\phi) \cos(\lambda)$ (Schmid's law) - slip begins when τ reaches τ_{crss}.
- True stress & strain: $\sigma_T = P/A_i$; $\varepsilon_T = \ln(L/L0)$; flow curve $\{r|\sigma = K \varepsilon^n\}$ (K = strength coefficient, n = strain hardening exponent).



Elastic: recoverable
Plastic: permanent (flow)
Strain hardening: stress rises as dislocations multiply & tangle
Necking: UTS (tension)

Fig 1.1 Stress-strain curve showing strain hardening

Strain hardening (work hardening)

As plastic strain increases, dislocation density rises (10^6 to 10^{12} $1/cm^2$); dislocations tangle and obstruct each other, so strength & hardness increase while ductility decreases. Cold working a metal: higher yield strength, hardness, brittleness, electrical resistivity increase; conductivity & corrosion resistance slightly fall; residual stresses and anisotropy appear.

2. Recovery, Recrystallization & Grain Growth

A cold-worked metal stores strain energy (about 10 % of the work). On heating (annealing) it moves back to the low-energy state in three overlapping stages:

Stage	What happens	Properties
Recovery (low temp, < 0.3 Tm)	Point defects annihilate; dislocations rearrange to low-energy cells (polygonisation); internal stresses relieved; no change of grain shape	Hardness / strength almost same; ductility slightly up; electrical conductivity recovers
Recrystallization (~0.3–0.5 Tm)	New strain-free equiaxed grains nucleate (at grain boundaries, slip bands) and grow, consuming deformed grains	Strength & hardness fall sharply; ductility restored; fully soft
Grain growth (higher T / longer time)	Large grains grow at the expense of small ones to reduce boundary energy	Strength falls further; coarse grain gives poor toughness & rough surface (orange peel)

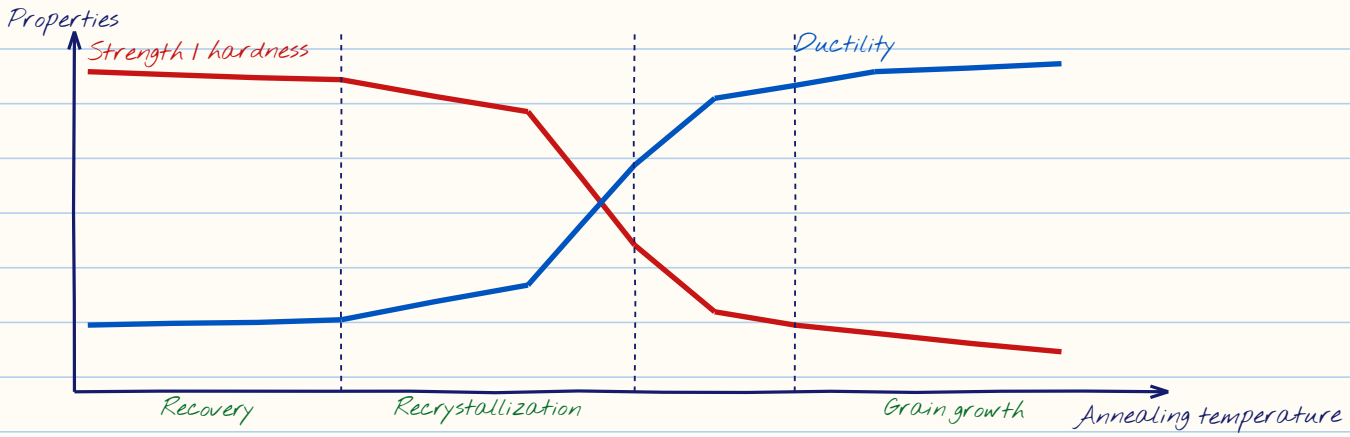


Fig 2.1 Effect of annealing on cold-worked metal

Recrystallization temperature (T_{rex})

- Lowest temperature at which complete recrystallization occurs in about 1 hour: about 0.3 to 0.5 of the absolute melting point (K). e.g. Fe ~ 450 C, Cu ~ 200 C, Al ~ 150 C, Pb & Sn - room temp.
- Lowers with: more prior cold work, finer initial grain, pure metals, longer time. Rises with alloying/impurities.
- Final grain size smaller if: more deformation, lower annealing temp, shorter time. Critical deformation (2-8 %) gives very coarse grains - avoid.

3. Hot Working & Cold Working

Aspect	Hot working	Cold working
Temperature	Above recrystallization temp (0.5-0.75 Tm)	Below recrystallization temp (room temp usually)
Force required	Low (flow stress is low)	High (strain hardening)
Strain hardening	None - recrystallizes continuously	Yes - strength, hardness up
Grain structure	Refined, equiaxed; blow holes closed; fibrous flow lines	Distorted, elongated grains; anisotropy
Surface finish & accuracy	Poor; scale formed, oxidation, tolerance +- 1-2 %	Good finish, close tolerance (+- 0.1 mm)
Ductility	High, large deformation in one step	Limited; intermediate annealing is needed
Residual stress	Nil	Present (cause distortion)
Applications	Forging, hot rolling, extrusion, hot piercing - ingots, large sections	Cold rolling sheet, wire drawing, cold heading, bolts
Cost	Heating & scale loss	Less energy but higher power & tooling

- Warm working: between the two (0.3-0.5 Tm) - reduces forces compared with cold, better finish than hot.
- Annealing: heating to a suitable temperature, holding, and slow furnace cooling to soften the metal, relieve stresses, refine grain, improve machinability & ductility. Types: full, process (recrystallization) annealing, stress-relief, spheroidising, normalising (air cooling).
- Hot-working effects: homogenises structure, breaks dendritic cast structure, heals porosity, refines grains, and creates fibre flow lines (mechanical fibering)

which give directional strength - e.g. crankshafts forged along the contour.

Effect of temperature & strain rate on flow stress

- Flow stress falls with rise in temperature; at hot-working temperature it is nearly independent of strain (no hardening) but depends on strain rate: sigma = C (strain rate)^m (m = strain-rate sensitivity: 0.02-0.4 hot, ~0.01 cold).
- High m (superplastic alloys, m > 0.5) resists necking -> huge elongations (500-2000 %) in superplastic forming.
- Hot shortness: some metals (S in steel) crack when hot worked - Mn added to form MnS. Cold shortness: brittleness at low temp due to P.

Solved example 1 - true strain

A bar 100 mm long is cold worked to 150 mm length. Engineering strain e = (150-100)/100 = 0.5; true strain epsilon = ln(150/100) = 0.405. If K = 500 MPa & n = 0.2, flow stress = 500 x $(0.405)^{0.2}$ = 500 x 0.834 = 417 MPa.

4. Forging

Forging is a bulk-forming process in which metal is shaped by compressive forces applied through hammers or presses (dies). Used for crankshafts, connecting rods, gears, turbine discs, spanners, bolts. Provides grain flow, high strength and toughness (better than castings).

Types of forging

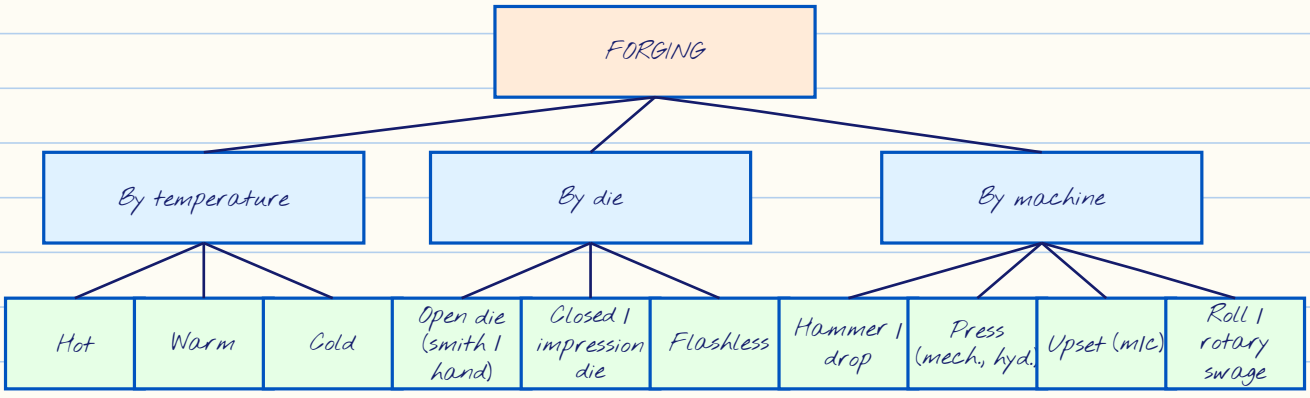


Fig 4.1 Classification of forging

Type	Description & uses
Open-die (smith) forging	Workpiece between flat/simple dies; metal flows freely. Operations: upsetting, drawing-out (cogging), fullering, edging, bending, punching, cutting. Heavy shafts, rings, discs.

Closed / impression-die (drop) forging	Heated billet in two shaped dies; excess metal flows out as flash (gutter) giving pressure to fill the cavity. Close tolerance, mass production: connecting rods, spanners, gears.
Upset forging	Horizontal forging machine; a bar is gripped and its end is upset (head formation): bolts, rivets, valve stems; cold heading.
Press forging	Slow squeezing action (hydraulic: 100–50000 t; mechanical: crank) – deep penetration, uniform flow, dies with less draft.
Roll forging	Bar passed through a pair of grooved segment rolls to reduce areal taper: axles, tapered levers, leaf springs.
Swaging / rotary	Dies hammer a rod rapidly while rotating – reduces diameter or makes tapers (tapered pins, tube ends).
Flashless / precision	Closed die with accurate billet volume; no flash; near net shape.

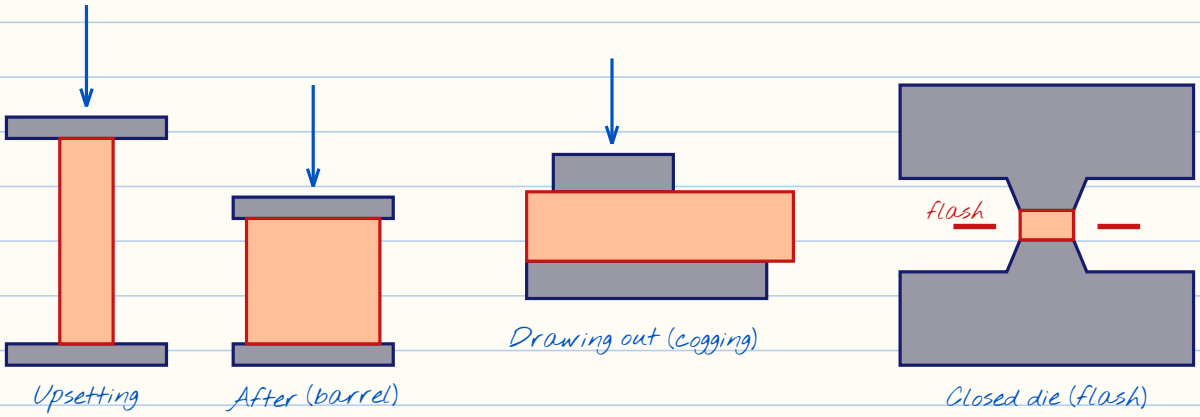


Fig 4.2 Forging operations

Steps in a typical drop-forging cycle

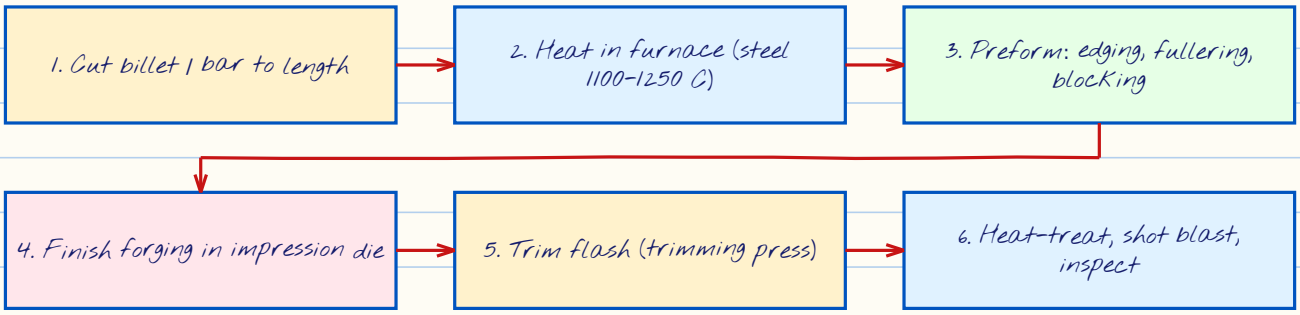


Fig 4.3 Forging sequence

- Heating: gas / oil-fired, induction or resistance heating; temperature ranges – Al 400–480 C, Cu alloys 700–900 C, carbon steel 1000–1250 C, Ti 800–1000 C. Overheating causes grain growth / burning; scale loss 1–3 %.
- Lubricants: graphite in oil / water, glass, MoS_2 – reduce friction & die wear and help release.

- Advantages of forging: high strength-to-weight, grain flow follows contour, no internal voids, good fatigue & impact strength, high reliability, good surface (cold forging). Limitations: high tooling cost, flash waste, limited to simple shapes without undercuts, scale & distortion in hot forging, poor tolerance.

Solved example 2 - upset forging load

A cylinder $d = 100$ mm, $h = 50$ mm is upset between flat dies; flow stress $\sigma_f = 200$ MPa, $\mu = 0.2$. Area $A = \pi/4 \times 100^2 = 7854$ mm^2. $F = \sigma_f A (1 + \mu d / 3h) = 200 \times 7854 \times (1 + 0.2 \times 100 / 150) = 1.5708 \times 10^6 \times 1.133 =$ 1.78 MN (about 178 tonnes).

Forging equipment & force

- Hammers: board drop (gravity), power (steam / air) - impact; energy $E = 0.5\, m\, v^2$. Presses: mechanical (crank / knuckle) - stroke limited; hydraulic - constant force.
- Forging load (open-die upsetting of cylinder): $F = \sigma_f \times A \times (1 + \mu d / 3h)$ (σ_f = flow stress, A = contact area, d = diameter, h = height, μ = friction coefficient). Friction increases pressure towards centre - friction hill.
- Closed-die load roughly $F = K \sigma_f A$, with $K = 3$-12 depending on shape complexity.
- Die design: parting line, draft (3-7 degrees), fillets & corner radii, flash gutter, material allowance, shrink allowance (about 1.2 %).

Forging defects & remedies

Defect	Cause	Remedy
Laps / folds (cold shut)	Metal folded over and pressed in without fusing; sharp die corners; wrong billet size	Generous fillets, correct preform and material flow
Incomplete die fill / unfilled section	Insufficient metal, cold die/billet, poor lubrication, low pressure	Proper billet volume & temp, adequate force, preheated die
Surface cracks (hot shortness)	Working below proper temp or sulphur-rich steel; excessive reduction	Heat to correct range, slow deformation, control S
Scale pits	Oxide scale pressed into surface	Descale before forging, protective atmosphere, lubricants
Die shift / mismatch	Misalignment of upper & lower dies	Align dies, lock with dowels / guides

Buckling	Long slender billet in upset (h/d > 3)	Limit L/d < 3, guide dies
Burning / overheating	Very high heating (coarse grain, brittle)	Control furnace temp
Flakes / internal bursts	Hydrogen & cooling stress	Slow cooling, anneal

5. Rolling

Rolling reduces thickness (or changes cross-section) of a long workpiece by compressive forces from two (or more) rotating rolls. It is the most widely used metal-working process (~90 % of all metal is rolled at least once). Hot rolling of ingots/billets -> blooms, billets, slabs; then finish rolling.

- Products: bloom (> 150 x 150 mm sq.), billet (40–150 mm sq.), slab (rectangular, > 200 mm wide), plates (> 6 mm), sheets (< 6 mm), strips, foils, rods, bars, rails, structural shapes (I, H, channels, angles), tubes (Mannesmann piercing), rings, threads.
- Flat rolling parameters: initial thickness h0, final hf; draft d = h0 − hf; roll radius R; contact length L = sqrt(R (h0 − hf)); reduction r = d/h0; strain epsilon = ln(h0/hf).
- Bite condition: roll grips if tan alpha <= mu; max draft = mu^2 R (alpha = angle of contact = sqrt(d/R)).
- Neutral (no-slip) point: where roll velocity = strip velocity; before it strip slower than roll (friction pulls in), after it faster (forward slip).
- Roll force F = L x w x sigma_avg (w = width, sigma_avg = average flow stress in the bite). Torque T = F x (L/2) per roll (arm approximately 0.5 L); Power P = 2 pi N T x 2 rolls = 2 x 2 pi N F L/2 W = 2 pi N F L (N in rev/s).

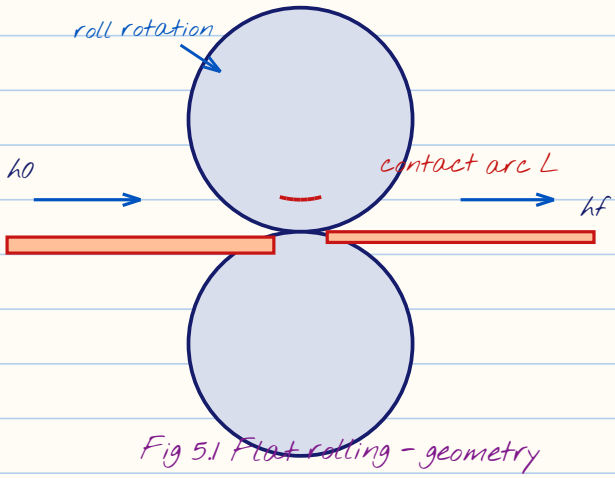


Fig 5.1 Flat rolling - geometry

Neutral point: v(roll) = v(strip)

Entry: strip slower than roll

Exit: strip faster (forward slip)

F = L w sigma_avg

Power = 2 pi N F L

Types of rolling mills

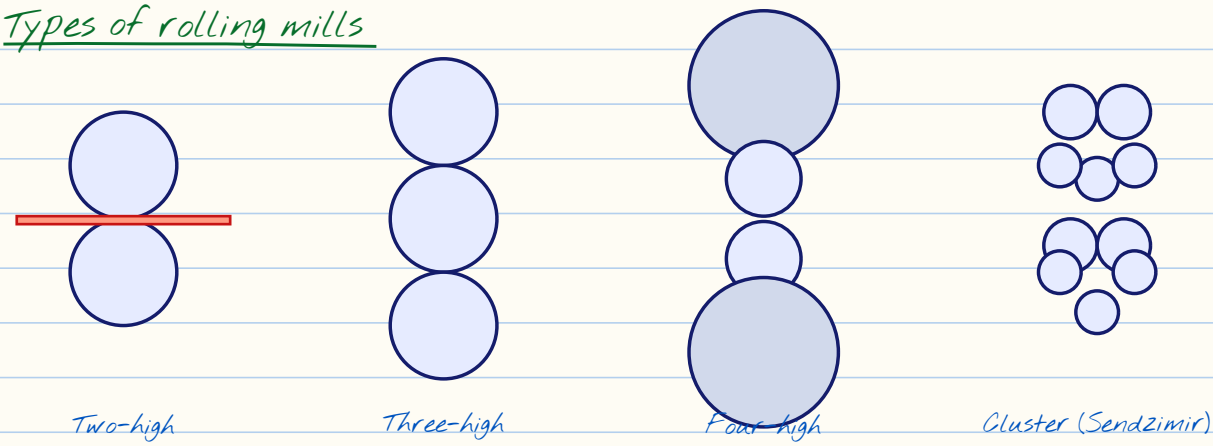


Fig 5.2 Rolling mill arrangements

Mill	Features & use
Two-high (reversing / non-reversing)	Simple; two rolls same size. Reversing - hot rolling blooming, slabbing, plates, rails. Non-reversing - continuous finishing.
Three-high	Upper, middle, lower rolls; the work passes through between alternate pairs; no reversal needed; for structural sections, rails (billets).
Four-high	Two small work rolls backed by two large backup rolls -> reduces roll deflection & force; for plates, sheets, strip (hot & cold).
Cluster (Sendzimir)	Small work rolls supported by many backup rolls - thin foils / hard stainless strip.
Tandem (continuous)	Series of stands, strip simultaneously in several - high speed strip production (hot strip mill / cold mill).
Planetary & ring rolling	Two heavy backup rolls with many small planetary rolls - 90 % reduction in one pass; ring mills for bearing races, wheel rims.

Hot rolling	Cold rolling
Done above T_rex; ingot -> bloom / slab / billet; large reductions (20-50 % per pass)	Done at room temp on hot-rolled, pickled strip; small reductions (< 30 %)
Rough surface, scale, tolerance 2-5 %; thickness above 2 mm	Bright finish, close tolerance (+- 0.01 mm), thin sheets, foils
No work hardening; low roll force	Work hardened (temper-rolled); higher force; roll flattening
Products: plates, rails, I-beams, bars	Products: car-body sheet, tinplate, strip, foils

Seamless tube production (Mannesmann piercing)

- Shaped products: roll-pass design - bar passes through successive grooves (box,

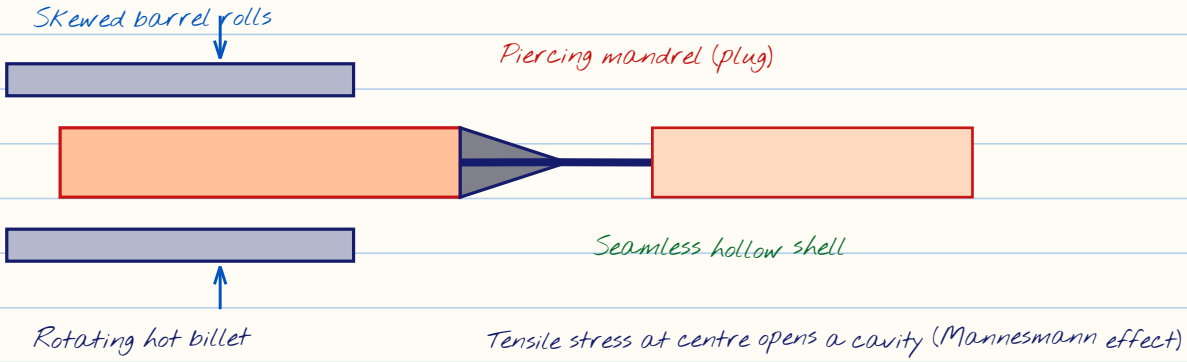


Fig 5.3 Rotary piercing of a hot billet

diamond, oval, round) to give rods / angles / rails; spread is the increase in width (0.35 x draft).

- Rolling of rings, threads (cold, thread rolling dies - 10x faster than cutting, stronger), gears, shot peening.
- Controls: roll gap, roll speed, tension (front / back), lubrication (cold - emulsion), roll cooling water.

Solved example 3 - flat rolling

Plate 25 mm thick is rolled to 20 mm with rolls of $R = 250$ mm, width $w = 300$ mm, $\sigma_{avg} = 250$ MPa, $N = 100$ rpm, $\mu = 0.1$.

- Draft = 5 mm; max draft = $\mu^2 R = 0.01 \times 250 = 2.5$ mm per pass => cannot be done in one pass by friction alone; two passes required (or lubricant removed).
- Contact length $L = \sqrt{250 \times 5} = 35.4$ mm. Roll force $F = L w \sigma = 35.4 \times 300 \times 250 = 2.65$ MN.
- Power $P = 2\pi N F L = 2\pi (1.667)(2.65 \times 10^6)(0.0354) =$ about 982 KW.

Rolling defects

- Wavy edges: rolls bend (deflection) -> edges elongate more; cure - roll crowning, back-up rolls, bending.
- Zipper / centre cracks, edge cracks: low ductility, excessive reduction; Alligatoring: ingot splits in two tongues (non-uniform deformation); Residual stress / camber, scale, laps.

6. Extrusion

Extrusion forces a billet through a die opening by a ram, producing a long section of constant cross-section: rods, tubes, I/T/L shapes, window frames, aluminium profiles. Extrusion ratio $R = A_0 / A_f$ (typically 10-100). Billet is hot (Al: 400-500 C; Cu: 700-900 C; steel 1100-1250 C) or cold, with lubricants (glass, graphite).

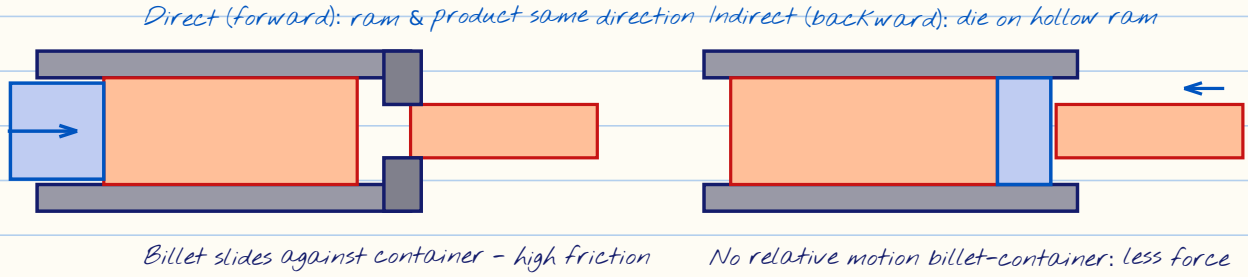


Fig 6.1 Direct and indirect extrusion

- Direct: most common; ram pushes billet through die at the opposite end; dummy block; friction with container -> higher force (+25-30 %), butt remains (10 % scrap).
- Indirect (backward): die at the ram end moves; product comes out through the hollow ram; billet static - friction absent, 25-30 % lower force; but ram support limits length.
- Hydrostatic: billet surrounded by pressurised fluid (castor oil, 1400 MPa) - no billet-container friction; brittle materials (Mo, W, ceramics) extruded; high reduction (up to 100:1); costly. Clad wires also.
- Impact extrusion: cold extrusion with a fast punch hitting slug placed in die: product moves forwards|backwards|combined - thin-walled tubes: toothpaste tubes, aluminium cans, battery cases, aerosol containers; Al, Pb, Sn, Cu, Mg slugs.
- Tube extrusion: mandrel with the ram forms hollow shapes; seamless tubes also by piercing + extrusion.
- Extrusion force: $F = A0 \times K \times \ln(A0/Af)$ where K = extrusion constant (MPa; Al 250 hot, steel 600+) (plus friction & redundant work terms).

Solved example 4 - extrusion

Billet dia 100 mm is extruded to a round rod 40 mm dia (Al, K = 250 MPa). Extrusion ratio $R = (100/40)^2 = 6.25$; $\ln R = 1.833$; $A0 = 7854\ mm^2$. Ram force $F = A0\ K \ln R = 7854 \times 250 \times 1.833 = 3.6\ MN$ (360 t). Die angle 90 degrees (flat) or conical (30-60) reduces dead-metal zone.

- Dies: flat (square) die - dead metal zone, good surface; conical|streamlined - lower force; porthole | bridge dies for hollow sections (aluminium window frames); die steel H13, tungsten carbide.
- Lubrication: hot - glass powder, graphite (Ugine-Sejournet for steel); cold -

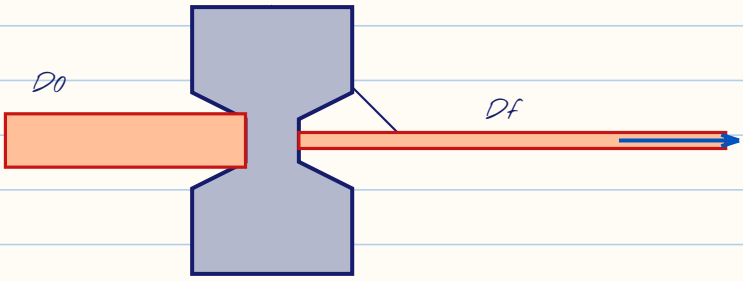
phosphate coat + soap.

Extrusion characteristics, defects

- Advantages: complex cross-sections, high reduction in one step, good surface & dimensional accuracy, no waste; seamless tubes; compressive stress state allows brittle metals.
- Defects: centre-burst (chevron cracking) due to tensile stress at centre (small die angle, low reduction); piping (tail-pipe) - surface oxide drawn to centre at the end; surface cracks (fir-tree) - speed/temperature too high; also die wear, shape distortion.

7. Wire Drawing & Tube Drawing

Drawing reduces the cross-section of a rod/wire/tube by pulling it through a tapered die. Unlike extrusion, material is pulled (tensile at exit) and compressed laterally. Mostly cold drawing - excellent finish, tolerance and strength. Wire diameter 0.025-12 mm; drawing speed up to 50 m/s.



Die: approach (alpha 6-10 degrees), bearing land, back relief; WC / diamond

Draw stress = sigma_avg ln(A0/Af)

Reduction per pass 20-40 %

Annealing between passes

Fig 7.1 Wire drawing through a die

- Drawing stress sigma_d = sigma_avg ln(A0/Af) (ideal, frictionless, homogeneous); with friction & redundant work about 1.2-2x. Maximum possible reduction per pass: about 63 % (when sigma_d equals flow stress) - practical 20-50 %.
- Process: pointing, pickling (descaling), coating (lime, phosphate, borax), lubrication (soap powder / oil), draw, repeated multi-die (continuous, tandem) with intermediate annealing, final stress relief.
- Die materials: tungsten carbide (steel), diamond (fine wires), tool steel. Lubricants: soaps, grease, oils.
- Defects: centre-burst, seams, die-marks, surface scratches, breaks; wire cold-worked becomes harder: tensile strength up to 2500 MPa (piano wire).

Tube drawing

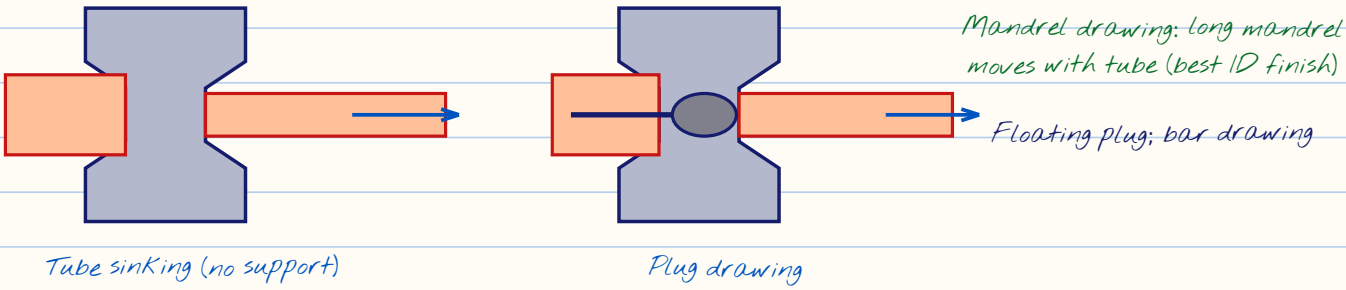


Fig 7.2 Tube drawing methods

- Tube sinking: only OD reduced; wall thickens; ID rough. Plug drawing: fixed/floating plug controls ID & thickness. Mandrel drawing: long mandrel supports the tube; draws with the tube - accurate ID, for long tubes.
- Used for seamless pipes, hypodermic needles, heat-exchanger tubes, bicycle frames; hot-extruded/pierced tubes are then cold drawn for accuracy.

Solved example 5 - wire drawing

A 10 mm wire is drawn to 8 mm; sigma_avg = 300 MPa. Area ratio A0/Af = $(10/8)^2$ = 1.5625; ln = 0.446; reduction = 1 - 0.64 = 36 %. Ideal draw stress = 300 x 0.446 = 134 MPa (< flow stress of exit wire, so wire will not break). Max reduction (ideal) = 1 - 1/e = 63 %.

Formula sheet - Unit III

Process	Formula	Terms
True strain / stress	epsilon = ln(L/L0); sigma_T = sigma_e (1 + e)	e = engineering strain
Flow curve	sigma = K epsilon^n	n = strain-hardening exponent
Recrystallization	T_rex = 0.3 to 0.5 Tm (K)	Tm = melting temperature
Upset forging	F = sigma_f A (1 + mu d / 3h)	A = contact area
Rolling	L = sqrt(R dh); F = L w sigma; P = 2 pi N F L; max dh = mu^2 R	dh = draft, N = revs
Extrusion	R = A0/Af; F = A0 K ln R	K = extrusion constant
Drawing	sigma_d = sigma_avg ln(A0/Af); max reduction = 63 %	ideal, no friction

Quick viva / 2-mark questions

- Define strain hardening. -> Increase in strength & hardness due to plastic deformation (dislocation pile-up).
- Why is hot working done above T_rex? -> To get continuous recrystallization, no hardening, low force.
- What is flash in forging? -> Excess metal squeezed out between dies - helps fill cavity & is trimmed.
- Why back-up rolls? -> To prevent deflection of small work rolls. What is the neutral point in rolling?
- What is extrusion ratio? -> A0/Af. What is piping? -> Oxide & impurities drawn in centre, tail-end defect.
- Difference between wire drawing & extrusion: pulling vs pushing; tensile vs compressive.

Important exam questions - Unit III

- Explain recovery, recrystallization & grain growth. Compare hot working & cold working. (10 M)
- Strain hardening and annealing. (5 M)
- Types of forging; forging defects & remedies. (10 M)
- Rolling: principle, types of rolling mills, roll force & power derivation, defects. (10 M)
- Types of extrusion; impact & hydrostatic extrusion; extrusion defects. (10 M)
- Wire drawing and tube drawing with neat sketches. (10 M)

Subject: Manufacturing Processes (23A0340IT)

UNIT - IV : SHEET METAL FORMING

Blanking & piercing, forces, deep drawing, stretch forming, bending & spring back, coining, spinning, presses & press tools, high energy rate forming (explosive, ~~electromagnetic, electro-hydraulic, rubber pad~~)

1. Introduction to Sheet Metal Forming

Sheet metal working deals with thin plates (thickness 0.4 – 6 mm) shaped on presses using punch and die. Surface-area-to-volume ratio is high; the thickness remains nearly constant (except shearing). Products: car bodies, aircraft skins, utensils, cans, enclosures, appliances, electrical laminations.

- Operations are of three kinds: cutting (shearing) – blanking, piercing, notching, trimming, parting; bending; drawing / forming – deep drawing, stretching, spinning, coining, embossing.
- Sheet materials: low-carbon steel, stainless steel, Al alloys, brass, copper, Ti. Required: high ductility, high n, high plastic strain ratio r (anisotropy).

2. Blanking & Piercing

Both are shearing operations: sheet is sheared between a punch and a die having a small clearance.

- Blanking: punching out a piece (the blank) from the sheet – blank is the product, remaining strip is scrap.
- Piercing: making holes – punched slug is scrap, the sheet with hole is the product.
- Other: notching (cutting from edges), lancing, trimming (removing flash), shaving (fine finishing cut), fine blanking (very smooth edge, triple-action press), parting (separating).

Clearance, forces & power (important formulae)

- Clearance c (per side) = 0.05 to 0.10 t (soft/ thin ~ 3–5 % t; hard ~ 8–10 % t); or c = 0.0032 t sqrt(tau). Blanking: die = blank size, punch smaller by 2c. Piercing: punch = hole size, die larger by 2c.
- Too small clearance: more force, secondary shear, rapid wear. Too large: burrs, rounded edge, rollover.
- Shearing (cutting) force $F = L \times t \times \tau_s$ (L = perimeter of cut, t = thickness,

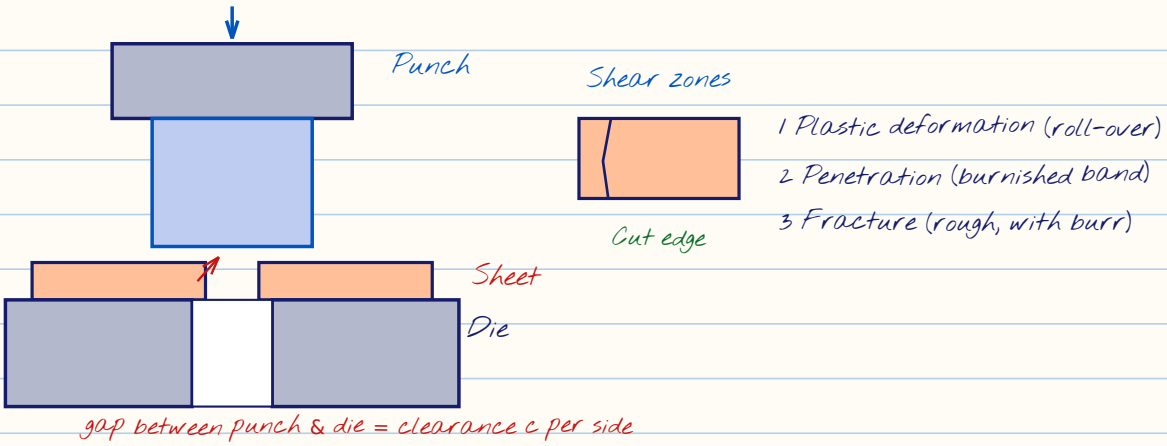


Fig 2.1 Blanking / piercing die set & cut-edge zones

tau_s = shear strength ~ 0.7 x UTS). For round hole: F = pi d t tau.

- Work done W = F x t x K (K = 0.3–0.6 fraction of t for penetration). Power = W x strokes/min.
- Stripping force = 2.5–10 % of cutting force (for hold-down). Press capacity = 1.2 x F.

Reducing cutting force: shear on punch / die

- Grinding a shear (bevel) angle on the punch or die face (shear = about t to 2t) makes cutting progressive - the force falls to 30–60 %, but the stroke increases. For blanking put the shear on the die so the blank stays flat; for piercing put it on the punch so the sheet (product) stays flat.
- Staggering punches (different lengths) in multiple-hole dies also reduces peak load.

Solved example 1 - blanking

Blank 50 mm dia, t = 2 mm, UTS = 400 MPa, tau = 0.7 x 400 = 280 MPa. F = pi x 50 x 2 x 280 = 87.96 KN (~ 8.97 t). With clearance 0.075 t = 0.15 mm/side: die size = 50 mm; punch = 50 - 0.3 = 49.7 mm. Press selected = 1.2 x 88 = about 106 KN.

Sheet-metal cutting operations at a glance

Operation	Description	Product / scrap
Blanking	Cut closed outline from strip	Blank is product
Piercing / punching	Hole made by punch	Slug is scrap
Notching	Removal of metal from edge of sheet	Notched part
Lancing	Cut and bent in one stroke, no metal removed (louvres, tabs)	Part with tab

Slitting / parting	Straight cut separating strip; two cut-offs	Two pieces
Trimming / shaving	Removing flash or excess; shaving gives smooth edge (0.1–0.2 mm)	Finished edge
Fine blanking	V-ring impingement, small clearance (0.5 %); fully burnished edges	Gears, cams

Strip layout & economy

- Strip layout aims to minimise scrap: nest blanks (staggered rows), provide bridge (web) between blanks and edge margin of about 1–1.5 t (min 1.5 mm).
- Material utilisation = (area of blanks per stroke / area of strip fed per stroke) × 100 – target above 70–75 %.
- Scrap often reused for small parts. Die life 10^5 – 10^6 strokes; resharpening by grinding.

3. Bending

Bending deforms the sheet about a straight axis (neutral axis): the outside is in tension, inside in compression; the neutral axis shifts toward the inner side (K factor 0.33–0.5).

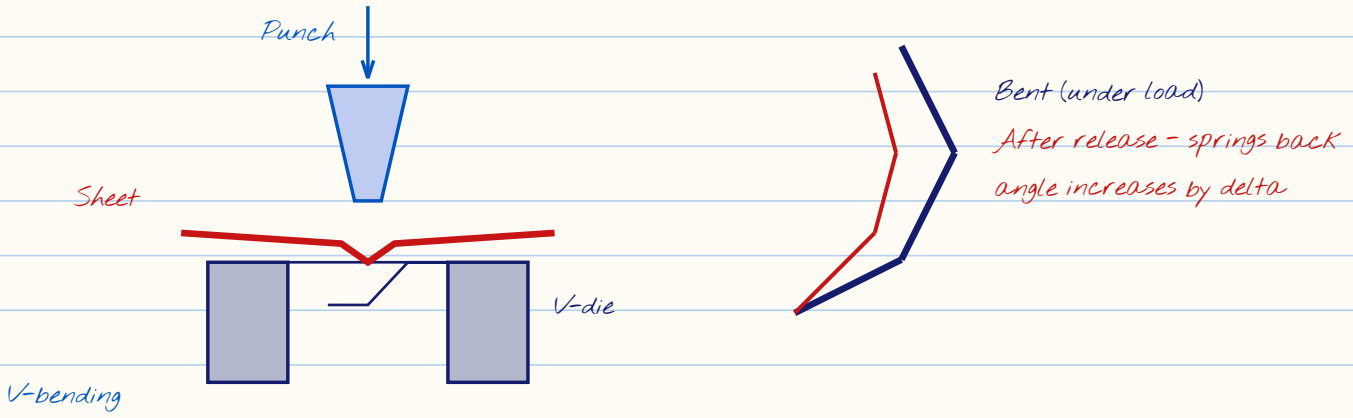


Fig 3.1 V-bending and spring-back

- Types: V-bending, edge (wiping) bending, U-bending, channel, curling, flanging, hemming, seaming, roll bending, air bending, bottoming, coining.
- Bend allowance (length of neutral axis): $BA = \alpha (R + K t)$ (α in radians, R = inner radius, K = 0.33 for $R < 2t$, 0.5 for $R > 2t$). Developed length = sum of straight lengths + BA.
- Minimum bend radius: $R_{min} \sim (0.5 \text{ to } 4)\, t$ – smaller for ductile; bend across rolling direction to avoid cracking.
- Bending force: $F = K_f \, UTS \, L \, t^2 / W$ (L = length of bend, W = die opening ~ 8 t,

K_f = 1.33 for V-die, 0.33 for wiping/ edge).

Spring back & remedies

After removal of load, the elastic recovery makes the bend angle open slightly - spring-back. It increases with higher yield strength, larger R/t, smaller thickness, and large die opening. Ratio $R_i/R_f = 4(R_i \sigma_y / E t)^3 - 3(R_i \sigma_y / E t) + 1$.

- Over-bending: bend more than needed so it springs to the correct angle (compensating angle).
- Bottoming / coining the bend: apply high pressure at end of stroke so that the sheet is plastically set.
- Stretch bending (apply tension while bending), ironing the bend zone, hot / warm bending.
- Use smaller die opening, lower R/t, punch with relief (spring-forming dies), and select low yield-strength / high-E material.

Solved example 3 - bending

Bend a 3 mm thick, 100 mm wide strip through 90 degrees with inner radius 6 mm (R = 2t so K = 0.33-0.5; take 0.4). $BA = (\pi/2)(6 + 0.4 \times 3) = 1.571 \times 7.2 = 11.3$ mm.

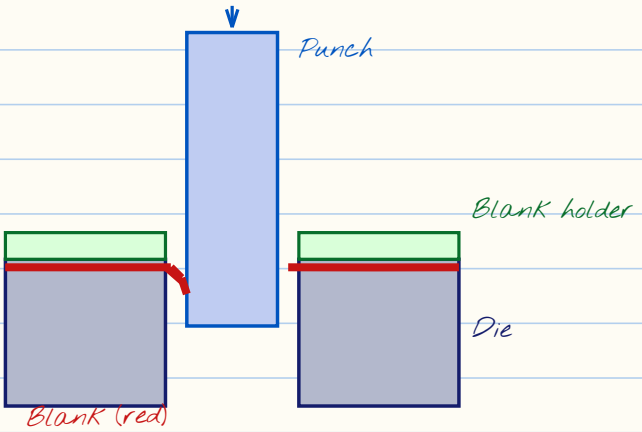
V-die opening W = 8t = 24 mm, UTS = 400 MPa, K_f = 1.33: $F = 1.33 \times 400 \times 100 \times 3^2 / 24 = 19.95$ KN (about 2 t).

Roll forming, tube bending & hydroforming

- Roll forming: strip passes through successive roll stands (6-20) that progressively bend it into long channels, roof sheets, door frames, tubes - continuous, fast.
- Tube bending: rotary draw bending with mandrel to prevent flattening; bend radius >= 1.5 d.
- Hydroforming: fluid pressure (up to 100 MPa) pushes a tube or sheet against a die - complex hollow auto parts (chassis rails).
- Superplastic forming: Ti / Al sheet at 900/1500 C stretched with gas pressure into a die (elongation > 300 %).

4. Deep Drawing

Deep drawing converts a flat blank into a hollow cup / box by pulling it into a die cavity with a punch; the flange is drawn inward (radial tension, circumferential compression). A blank holder presses the flange to prevent wrinkling. Examples: Kitchen sinks, pans, cans, shells, fuel tank halves.



DR = D / d (draw ratio); limiting DR ~ 2

First draw reduction 40–50 %

Radii: die = 4–10 t, punch = 4–10 t

Clearance = 1.1 – 1.2 t

Blank dia: $D = \sqrt{(d^2 + 4 d h)}$

Draw force $F = \pi d t \cdot UTS (D/d - 0.7)$

Fig 4.1 Deep drawing of a cup

- Blank diameter for a cylindrical cup of diameter d & height h (area equality): $D = \sqrt{(d^2 + 4 d h)}$.
- Drawing ratio / limiting draw ratio (LDR) = D/d_{max} ~ 1.8 – 2.2. Reduction r = (D – d)/D (max 0.5). Deeper cups need redrawing (reverse / direct, with inter-stage annealing); ratio of later draws 1.3–1.2.
- Drawing force: $F = \pi d t \cdot UTS (D/d - 0.7)$ approx; blank holding force = 1/3 of drawing force (pressure 1–3 MPa).
- Lubrication: oil, wax, polythene film, soap (reduces friction), die & punch polished.
- Ironing: further thinning of wall by passing through a clearance < t (cans). Drawing in double-action press: outer ram = blank holder, inner ram = punch.

Defect	Cause	Remedy
Wrinkling (flange / wall)	Low blank-holder force, thin sheet, large flange, die radius large	Increase holding force, draw beads, thicker sheet
Tearing / fracture at cup bottom	Too high draw ratio, sharp radii, high friction, low holder force	Increase radii, lubricate, anneal, reduce DR, redraw
Earing	Planar anisotropy of rolled sheet (variation in r)	Use sheet with low delta-r; trim ears
Surface scratches / orange peel	Dirty dies, coarse grain	Polish dies, fine-grain sheet

Stretcher strains (Luders bands)	Yield-point elongation in low-C steel	Temper-roll (skin pass) the sheet

Solved example 2 - deep drawing

Cup $d = 50$ mm, $h = 40$ mm. Blank $D = \sqrt{(50^2 + 4 \times 50 \times 40)} = \sqrt{(2500 + 8000)} = 102.5$ mm; $DR = 102.5/50 = 2.05$ (> 2): two draws are needed. First draw to $d1 = 102.5/1.8 \sim 57$ mm, second to 50 mm.

5. Stretch Forming, Coining & Spinning

Stretch forming

Sheet is gripped at its edges by jaws, stretched beyond yield (tension) and wrapped over a form block (die). Because of nearly uniform tension, spring-back & wrinkling are small. Used for large, shallow contoured panels - aircraft skins, wing leading edges, car roof panels, curved architectural panels.

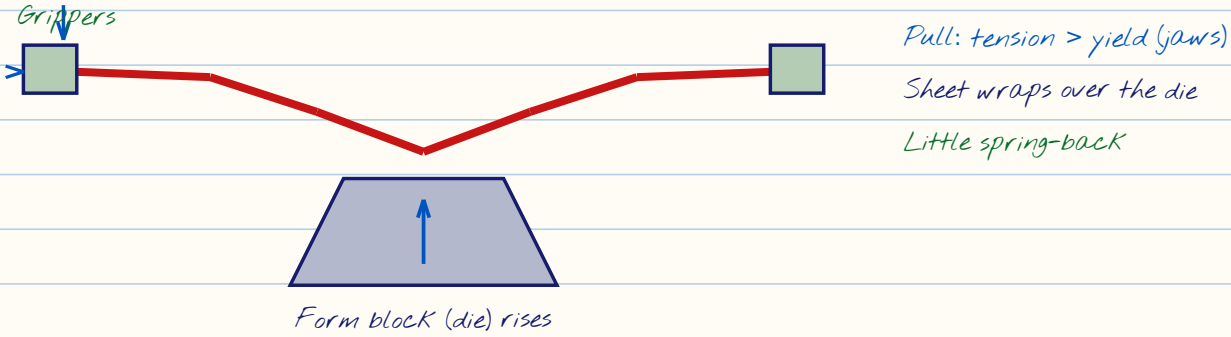


Fig 5.1 Stretch forming

Coining & embossing

- Coining: sheet / blank squeezed between closed dies under very high pressure (up to 5-6 times flow stress, 1600 MPa); metal flows to fill the finest details on both faces - no flash; thickness changes. Coins, medals, jewellery, hallmark, spring-back removal (bottoming).
- Embossing: shallow relief in a sheet by matching punch & die - thickness nearly the same (name plates, ribs). Coining force = $p \times A$ (p = 3-5 x UTS).

Spinning

A flat disc / tube clamped against a mandrel (former) on a lathe is rotated; a roller or tool progressively forces the metal over the mandrel. Products are axisymmetric: cones, hemispheres, reflectors, cooking vessels, rocket nose cones.

- Conventional (manual) spinning: wall thickness stays constant, diameter reduces - small quantities, cheap tooling.
- Shear spinning (power spinning): roller thins the wall: $t_f = t_0 \sin(\alpha)$ (sine law), diameter constant; accurate cones, no blank wrinkling. Tube spinning: reduces tube wall.
- Merits: low tooling cost, large parts, good finish, strain hardening. Limits: only symmetrical shapes, skilled labour.

6. Presses & Press Tools

A press is a machine tool with a ram (slide) reciprocating towards a fixed bolster - supplies force and energy for sheet operations. Capacity is in tonnes.

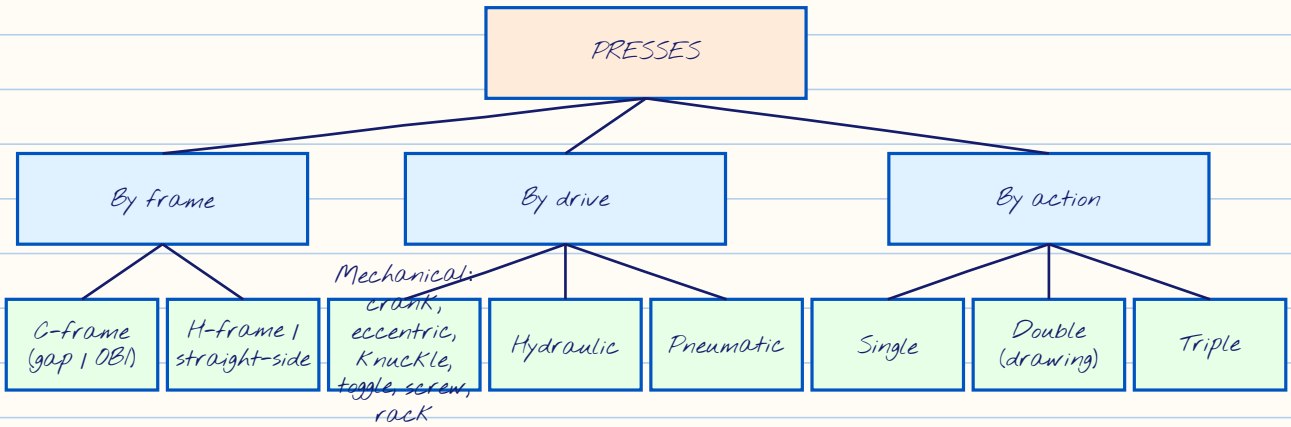


Fig 6.1 Classification of presses

Press	Characteristics & use
Mechanical (crank / eccentric)	Flywheel energy through clutch; fixed stroke; fast (20-1000 spm); blanking, piercing, shallow forming; force greatest at bottom dead centre.
Hydraulic	Oil pressure on piston; full force at any stroke, adjustable, slow - deep drawing, forming, forging, large parts; overload-safe.
Knuckle-joint / toggle	Huge force near end of stroke with slow ram speed: coining, embossing, sizing.
Screw press	Energy stored in flywheel & screw; used in forging, coining, minting.
Single / double / triple action	Double has blank-holder ram + punch ram (deep drawing); triple adds a bottom ram (reverse draw, ejector).

C-frame vs H-frame	C-frame - open front, easy access, lower rigidity (small work); H / straight-side - rigid, for heavy jobs & accuracy.

Press tools (dies)

- Simple (single-operation) die: one operation per stroke; low cost; low volume.
- Compound die: two or more cutting operations at one station in one stroke - e.g. blanking + piercing a washer; accurate concentricity, slow strip, parts fall back through die.
- Progressive die: strip moves through several stations; each stroke performs a different operation; finished part is cut off at the last station - high production, automatic.
- Combination die: cutting + non-cutting (blank + draw) in one stroke. Transfer die: separate blank moved by fingers between stations (large parts).

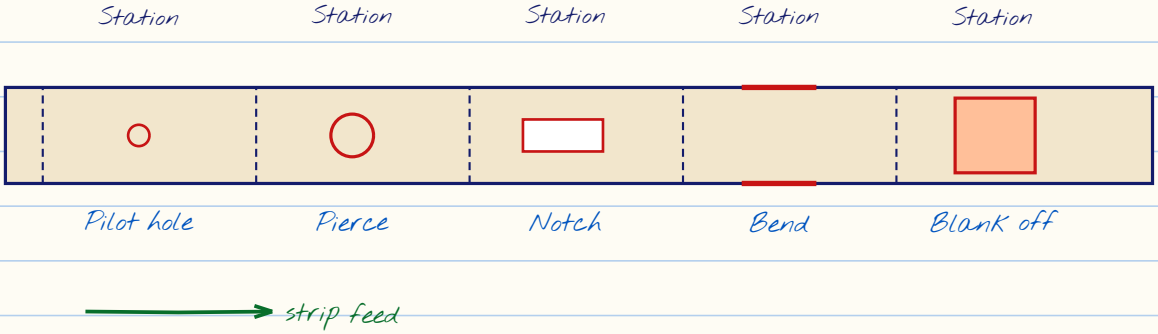


Fig 6.2 Progressive die - strip layout

Components of a press tool

- Punch & die, punch plate (holder), die block / shoe, stripper plate (strips sheet off the punch), guide pillars & bushes, pilot (locates strip), stops / gauges, shank, back-up plate, knock-out, blank-holder (draw dies), springs, fasteners & dowels.
- Materials: tool steel D2 / D3 (12 % Cr), A2, oil-hardening O1, high-carbon steel, tungsten carbide inserts; hardness 58-62 HRC. Die life improves with lubrication & correct clearance.

Formability of sheet metal

- Strain-hardening exponent n: high n -> better stretchability (uniform elongation). Plastic strain ratio r = width strain / thickness strain: high r (> 1) -> good deep drawing (resists thinning); average $r = (r0 + 2\,r45 + r90)/4$;

planar anisotropy delta-r = (r0 - 2 r45 + r90)/2 -> earing.

- Forming limit diagram (FLD): plot of major vs minor strain at the onset of necking - safe region below the curve.
- Tests: Erichsen cupping, Olsen, Swift cup, hydraulic bulge, limiting dome height (LDH).

Formula sheet - Unit IV

Item	Formula
Shearing force	F = L t tau (tau ~ 0.7 UTS); round: pi d t tau
Work / power	W = F x K t ; P = W x strokes per min
Clearance (per side)	c = 0.05 - 0.1 t
Bend allowance	BA = alpha (R + K t)
Bending force (V)	F = 1.33 UTS L t^2 / W
Blank diameter (cup)	D = sqrt(d^2 + 4 d h)
Drawing force	F = pi d t UTS (D/d - 0.7)
Draw ratio / reduction	DR = D/d (<= 2); r = 1 - d/D (<= 0.5)

Quick viva

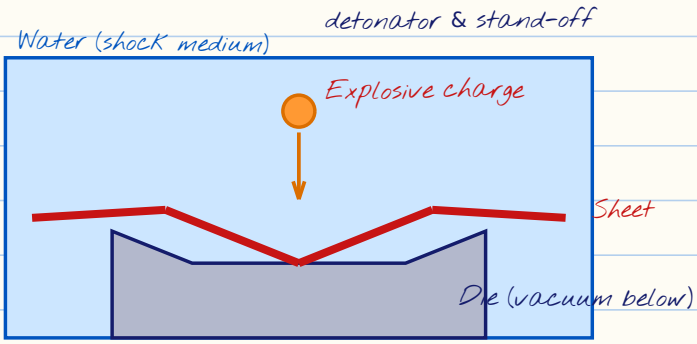
- Why shear on punch? -> To lower the peak cutting force. Why blank holder? -> To prevent wrinkling of flange.
- Which press for deep drawing? -> Double-acting / hydraulic press. Why coining needs a toggle press? -> Very high pressure at the end of stroke.
- What is spring-back? -> Elastic recovery of the bend after load is removed.

7. High Energy Rate Forming (HERF)

HERF processes release a large amount of energy in a very short time (milliseconds) to deform the metal at very high velocity (30-300 m/s). Useful for large, thin, hard-to-form materials (Ti, Ni alloys, stainless steel) with small production runs, where press cost would be high.

7.1 Explosive forming

Explosive charge (TNT, RDX, PETN, gun powder) is detonated in a medium (usually water) near the sheet; the shock wave / pressure pulse (up to 25 GPa) drives the sheet into the die. The die may be cheap (concrete, cast Zn/ Kirksite) with vacuum to pull the sheet.



Stand-off distance controls pressure

Large parts: ship plates, domes, tanks

Few parts, thick / large, low die cost

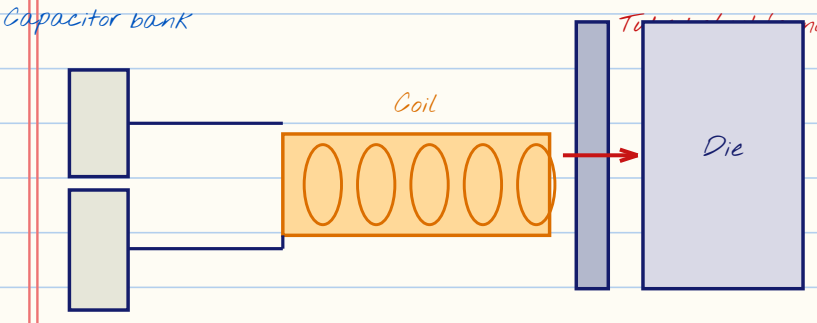
Safety & noise limits

Fig 7.1 Explosive forming (stand-off method)

- Stand-off: charge placed at distance in a medium (water) - uniform loading on the sheet. Contact: charge in contact with the work (for sizing/ cladding/ hardening). Confined: in a closed die; unconfined: open.
- Advantages: no costly press, very large parts, complex shapes, Ti & high-strength alloys, low tooling cost, less spring-back. Limitations: hazardous, skilled safe handling, low production rate, noise, licensing.

7.2 Electromagnetic forming (magnetoforming)

A capacitor bank is discharged through a coil placed near an electrically conductive work-piece; the rapidly changing magnetic field induces eddy currents in the work, which then interacts with the field (Lorentz force) and repels / pushes the workpiece at high speed against the die. Process time ~ 10-100 microseconds.



Used for: tube swaging, bulging, flanging, shallow drawing, assembly

Only conductive metals (Al, Cu, mild steel)

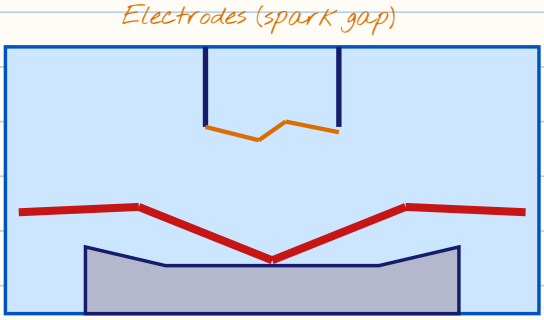
No contact tool, fast, repeatable

Fig 7.2 Electromagnetic (coil) forming

- Merits: contactless - no tool marks, high rate, good repeatability, no lubrication, less spring-back, can join dissimilar tubes (magnetic pulse welding). Limits: conductive materials only, coil life, work shape limited (tubular / shallow), small sheets, expensive power unit.

7.3 Electro-hydraulic forming (EHF)

Electric spark discharge (capacitors, 10-40 KV, 10-100 KJ) between two electrodes under water vaporises the fluid producing a high-pressure shock wave that forms the sheet into the die. Similar to explosive forming but easier to control and safe.

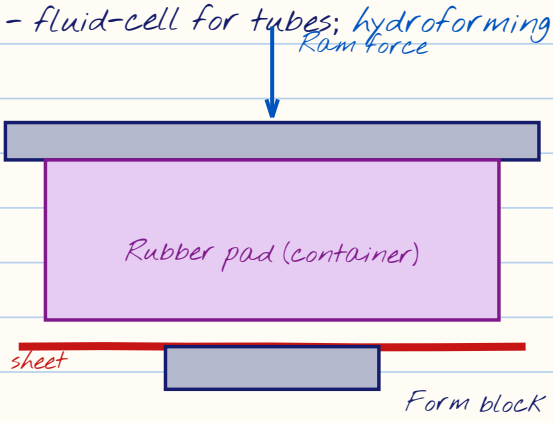


Spark vaporises water -> shock wave
No explosives; safer than explosive forming
Lower energy -> smaller parts, tubes, domes
Slower than EM; die vacuum used

Fig 7.3 Electro-hydraulic forming

7.4 Rubber pad forming

A rubber (polyurethane) pad replaces one half of the die. The sheet is pressed by the ram against a single form block; rubber behaves like a fluid (hydrostatic) pressing the sheet into the shape. Guerin process - rubber pad in a container, force up to 10 MPa+; Marform - rubber + blank holder for deep parts; Verson-Wheelon - fluid-cell for tubes; hydroforming - flexible diaphragm with fluid pressure.



Only one rigid die needed (cheap)
Shallow parts, brackets, aircraft ribs
No tool marks; many parts at once
Low pressure - small depth; pad wears

Fig 7.4 Rubber pad (Guerin) forming

Process	Energy source	Typical use	Advantage	Limitation
Explosive	Chemical explosive	Large domes, tanks, ship / aerospace	Huge parts, cheap die	Hazard, slow cycle
Electro-magnetic	Capacitor + coil	Tubes, shallow forms, assembly	Contactless, fast	Conductive only
Electro-hydraulic	Spark in water	Small/medium parts, tubes	Controlled, safe	Low energy, electrode wear

Rubber pad	Press + elastomer	Shallow sheet parts	Low tool cost	Limited depth, pad wear

General advantages & limitations of HERF

- Advantages: form difficult metals, less spring-back, little or no diel press cost, large parts, good accuracy & finish, strain-rate raises formability.
- Limitations: safety & training, noise, low production rate, limited shapes, energy control needed, special facilities (pits, tanks).

Important exam questions - Unit IV

- Differentiate blanking & piercing; clearance; calculate cutting force & power. (10 M)
- Deep drawing: explain with sketch; draw ratio; defects. Stretch forming & spinning (5 M).
- Bending: bend allowance, spring-back & remedies. Coining (5 M).
- Types of presses & press tools (compound, progressive...). (10 M)
- Explain explosive / electromagnetic / electro-hydraulic / rubber pad forming; advantages & limitations. (10 M each)

Subject: Manufacturing Processes (23A0340IT)

UNIT - V : ADDITIVE MANUFACTURING

Steps in AM, classification, advantages, materials, VAT photopolymerization, extrusion-based, powder bed fusion, directed energy deposition, post-processing, applications

1. Introduction to Additive Manufacturing (AM)

Additive manufacturing (3D printing, rapid prototyping, layered manufacturing) is a process of making a part by adding material layer by layer from a 3D CAD model, as opposed to subtractive (machining) and formative (casting, forging) processes. ASTM F2792 definition: "joining materials to make objects from 3D model data, usually layer upon layer".

- Evolution: Rapid Prototyping (1987 - first SLA machine by 3D Systems, Chuck Hull) -> Rapid Tooling -> Rapid Manufacturing -> Direct digital manufacturing of end-use parts.
- Principle: solid 3D model is sliced into thin 2D layers (0.01 - 0.5 mm); each layer is built by depositing / solidifying material and bonded to the previous layer; the part grows in the Z direction.
- Near-net shape; geometric complexity is free - internal channels, lattices, topology-optimised shapes.

	Subtractive	Formative	Additive
Principle	Material removed from block	Shape by deformation / mould	Material added layer by layer
Waste	High (chips)	Low-moderate	Very low (5-10 %)
Tooling	Cutters, fixtures	Dies, patterns (costly)	None
Complexity	Limited by tool access	Limited by die	Almost unlimited
Best volume	Low-medium	High	Low, customised

2. Steps in Additive Manufacturing

- Step 1 - CAD: solid or surface model; can come from CT/MRI or 3D scanner (medical, reverse engineering).
- Step 2 - STL: surface approximated by triangles (vertices + normals); resolution set by chordal tolerance; must be a closed (water-tight) manifold.

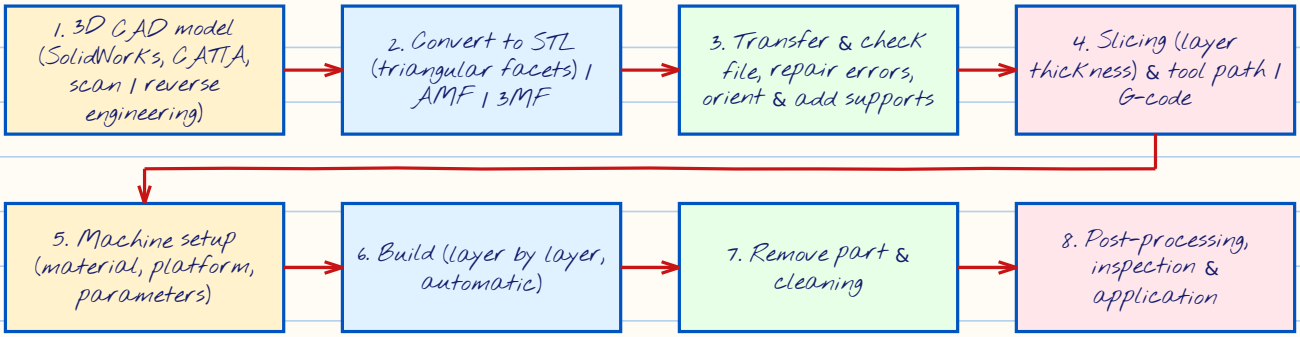


Fig 2.1 Generic AM process chain

Errors: gaps, flipped normals, overlapping triangles - repaired with Magics / Netfabb.

- Step 3 - Build orientation & supports: orientation affects build time, accuracy, strength (anisotropy: weakest in Z), surface finish and support quantity; supports hold overhangs (> 45 degrees) & resist warping.
- Step 4 - Slicing: STL cut into layers; slice thickness 20-100 microns (SLA) to 0.1-0.4 mm (FDM); produces contour & hatch paths. Staircase effect: stepped surface on curved faces - smaller layer = better finish but longer time.
- Step 5-6 - Setup & build: set temperature, laser power, scan speed, hatch spacing, layer thickness; build is automatic; build time from hours to days.
- Step 7-8: remove, clean, post-process (see section 8), inspect (CMM, CT scan), use.

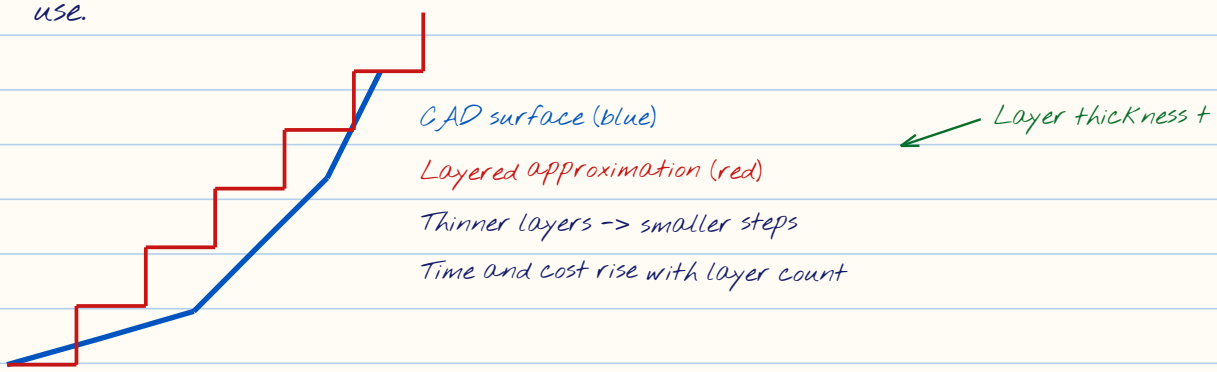


Fig 2.2 Staircase effect

3. Classification of AM Processes

ASTM F42 / ISO 17296-2 groups AM into seven process categories. Alternative classification: by starting material (liquid, powder, solid wire/sheet) or by energy source (laser, electron beam, thermal, UV).

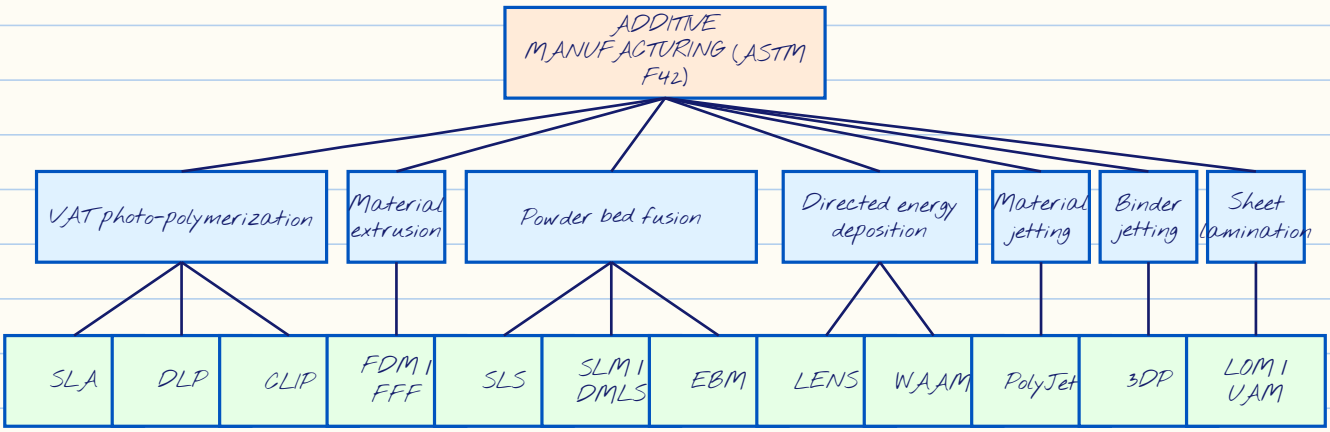


Fig 3.1 Classification of AM processes (7 categories)

Category	Energy / bonding	Material	Examples (companies)
Vat photo-polymerization	UV light cures liquid resin	Photopolymers, ceramics-filled resin	SLA (3D Systems), DLP, CLIP (Carbon)
Material extrusion	Heated filament extruded; thermal bonding	ABS, PLA, PC, PEEK, nylon	FDM (Stratasys), FFF desktop printers
Powder bed fusion	Laser / e-beam melts or sinters powder	Polymers, metals, ceramics	SLS, SLM, DMLS (EOS), EBM (Arcam)
Directed energy deposition	Focused thermal energy melts wire/powder as deposited	Metals (Ti, Ni, steels)	LENS (Optomec), WAAM, EBAM (SciaKy)
Material jetting	Droplets jetted & UV-cured	Photopolymers, wax	PolyJet, MultiJet (3D Systems)
Binder jetting	Liquid binder glues powder	Sand, metal, ceramic, gypsum	ExOne, Voxeljet, 3DP (Z Corp)
Sheet lamination	Sheets bonded (adhesive/ ultrasonic) & cut	Paper, plastic, metal foils	LOM (Helisys), UAM (Fabrisonic)

4. Advantages, Limitations & Materials

Advantages of AM

- Design freedom: complex geometry, internal channels, conformal cooling, lattices, consolidation of assemblies into one part (GE fuel nozzle: 20 parts -> 1).
- No tooling / dies - short lead time, quick design iteration, rapid prototyping,

economical for low volume.

- Mass customisation: patient-specific implants, hearing aids, dental aligners; each part can be different at no extra cost.
- Less material waste, lightweight topology-optimised parts, on-demand & distributed production (digital inventory).
- Multi-material and functionally graded parts possible; repair of expensive parts.

Limitations

- Slow build rate; part cost high for large volumes; size limited by build envelope.
- Surface finish (staircase), dimensional accuracy (+- 0.05 - 0.2 mm), anisotropy, porosity, residual stresses; limited range of qualified materials; need post-processing.
- Machine & material cost high (metal AM > 1 crore Rs.), skilled operators, safety (powders, lasers); certification & standards still developing.

Materials for AM

Class	Materials	Processes
Photopolymers (liquid resins)	Acrylates, epoxies; filled with ceramic / glass	SLA, DLP, CLIP, PolyJet
Thermoplastics	ABS, PLA, PC, PETG, nylon (PA12), PEEK, ULTEM, TPU	FDM, SLS
Metals	Ti-6Al-4V, stainless steel 316L / 17-4PH, Inconel 625/718, AlSi10Mg, Co-Cr, maraging steel, Cu, tool steels	SLM / DMLS, EBM, DED, binder jetting
Ceramics	Alumina, zirconia, silica, SiC, glass, bio-ceramics (HA)	SLA (filled), binder jetting, SLS
Composites	Carbon / glass fibre reinforced nylon, metal-polymer filaments	FDM, SLS
Others	Wax (investment patterns), sand (moulds), paper, food (chocolate), concrete, bio-ink (cells)	Binder jet, extrusion, LOM

- Powder requirements: spherical (15-45 microns) for flowability, good packing, low oxygen, narrow size distribution (gas atomised); filament: precise 1.75 / 2.85

mm dia.; resin: viscosity 50-1500 mPa s, photo-initiator.

5. VAT Photo-polymerization AM Processes

A vat of liquid photosensitive resin (photopolymer) is selectively cured by UV light / laser; monomers cross-link to a solid polymer. First commercial AM: Stereolithography (SLA).

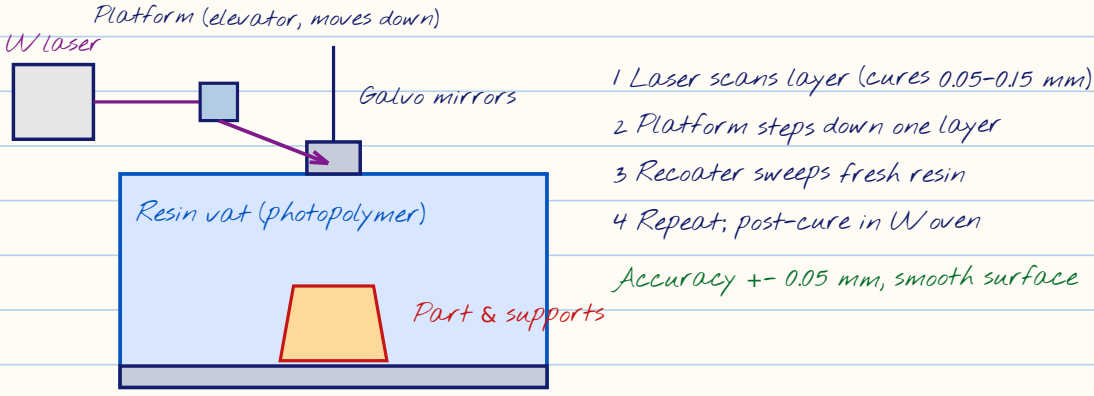


Fig 5.1 Stereolithography (SLA)

- SLA: UV laser (355 nm) draws each layer point-by-point (vector scan); platform lowers; supports are needed; part cured further in a UV oven. Layer 25-150 microns; build 250 - 2000 mm.
- DLP (Digital Light Processing): a projector cures the entire layer at once (pixel/voxel mask) - faster than SLA; resolution depends on pixels (e.g. 50 microns). LCD / MSLA: LCD mask + LED.
- CLIP / CLIP (continuous): oxygen-permeable window creates a "dead zone" so the part is pulled up continuously - 25-100x faster, smooth, isotropic parts. Bottom-up vs top-down: light from below (peel forces) / above (large vats).
- Process parameters: laser power, scan speed, layer thickness, hatch spacing, curing depth (Jacobs: $Cd = Dp \ln(E/Ec)$), resin viscosity, temperature.
- Advantages: best accuracy & surface finish among AM, fine details, large parts, transparent parts. Disadvantages: resin costly & toxic, brittle / UV-degrading parts, supports & post-curing needed, limited materials, messy. Uses: master patterns for investment casting, dental models, hearing aids, jewellery, visual prototypes, microfluidics.

6. Material Extrusion Based AM (FDM / FFF)

Fused Deposition Modelling (FDM, Stratasys 1989; FFF in open-source RepRap): thermoplastic filament is heated to semi-liquid in a nozzle and extruded along the path, bonding to the layer below by thermal fusion and solidifying quickly.

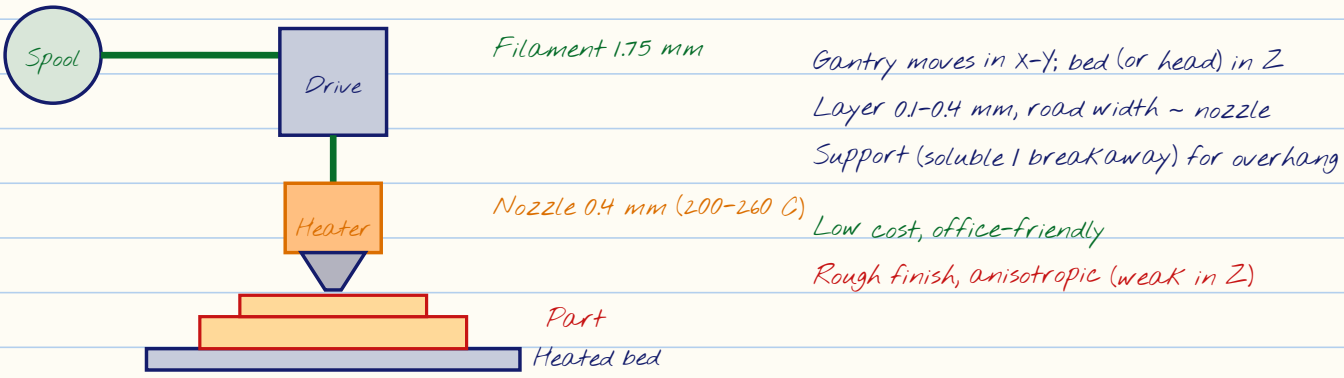
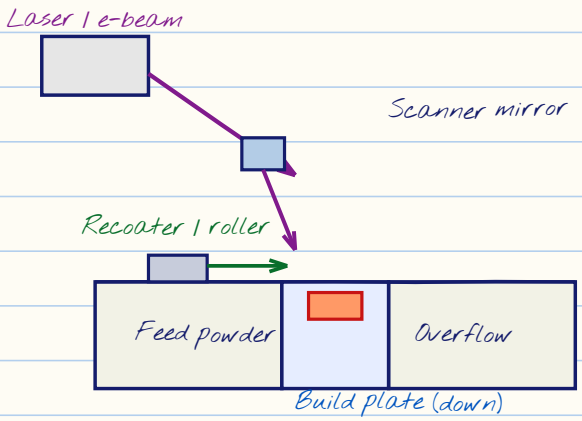


Fig 6.1 FDM / FFF process

- Parameters: nozzle & bed temperature (ABS 230/100 C; PLA 200/60 C), layer thickness, raster angle, air gap, infill % & pattern, print speed, number of contours, build orientation, cooling.
- Materials: PLA, ABS, PETG, nylon, PC, ULTEM / PEEK, TPU, carbon-fibre composites; water-soluble PVA / HIPS supports.
- Advantages: cheap machines & materials, simple, durable functional parts, multi-material, office use. Limitations: poor surface, slow for large parts, anisotropic strength (inter-layer bonds), warping/ shrinkage, visible layer lines, limited resolution.
- Defects: warping, stringing, under-extrusion, layer shifting, poor bed adhesion, voids between roads. Uses: prototypes, jigs & fixtures, tooling, drones, enclosures, education, custom prosthetics, robotics; also concrete / food / bio-printing (extrusion).

7. Powder Bed Fusion (PBF) AM Processes

PBF: a thin layer of powder is spread on the build platform by a recoater (blade / roller); a laser or electron beam selectively sinters / melts the powder along the layer contour; platform lowers; fresh powder is spread; repeat. Unfused powder supports the part (for polymers) and is reusable.



SLS: polymer powder (PA12), no support
SLM / DMLS: metal fully melted, supports
EBM: e-beam, vacuum, 700 C preheat
Inert gas (Ar / N2) for metals
Layer 20-100 microns (metal)

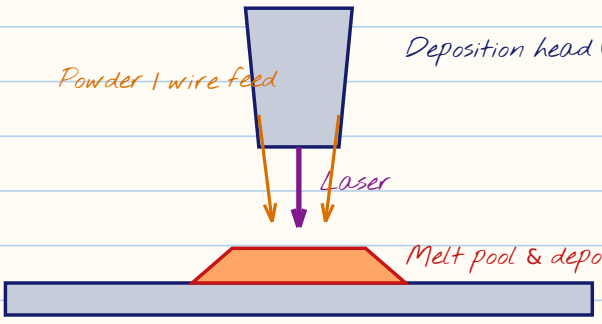
Fig 7.1 Powder bed fusion

Process	Energy / bonding	Features
SLS (Selective Laser Sintering)	CO2 laser (30-50 W) sinters polymer particles below melting (partial fusion)	PA12, TPU; no supports (powder supports); parts 95 % dense; porous
SLM / DMLS / LPBF	Fibre laser (200-1000 W) fully melts metal powder; inert Ar; supports needed	99.9 % dense; Ti, Inconel, SS, Al; heat treatment / HIP; residual stress
EBM (Electron Beam Melting)	E-beam (3-6 KW) in vacuum; bed pre-heated (~700 C+)	Less residual stress, no supports needed as much; Ti-6Al-4V, Co-Cr; rougher
Binder jet sintering (for contrast)	Binder + furnace sinter	No thermal stress, large volume, shrinkage

- Energy density: $E = P / (v \times h \times t)$ J/mm3 (P = laser power, v = scan speed, h = hatch spacing, t = layer thickness) - low E: lack of fusion / porosity; high E: Keyholing, balling, evaporation.
- Advantages: high density, fine features, mechanical properties like wrought, wide material range, no tooling, lattice & hollow parts, powder recyclable (SLS 50 % refresh).
- Limitations: costly equipment, slow, rough surface (Ra 6-15 microns), residual stresses & distortion, supports & thermal management (metal), powder safety (explosive / reactive), need HIP & machining.
- Uses: aerospace brackets / fuel nozzles, turbine blades, orthopaedic implants (porous Ti), dental crowns, heat exchangers, tooling inserts with conformal cooling, motorsport.

8. Directed Energy Deposition (DED) AM Processes

DED: a focused thermal source (laser, electron beam, plasma or arc) creates a melt pool on the substrate while metal powder or wire is fed coaxially/ side-fed into it; material is deposited and fused as the head (or workpiece) moves. Not a powder bed - deposition happens where the energy is.



LENS, DMD, WAAM, EBAM, laser cladding
Deposition rate 0.5 - 10 Kg/h (WAAM)
Large parts (metres), repairs, cladding
Poor accuracy -> machining needed
Multi-material / graded structures

Fig 8.1 Directed energy deposition

- Types: laser-based (LENS - Laser Engineered Net Shaping, DMD), WAAM (wire + arc, cheap, high rate), EBAM (electron beam wire, vacuum), plasma deposition, cold-spray (solid-state).
- Materials: Ti, Ni superalloys, stainless steels, Al, Cu, tool steels, cermets - powder or wire.
- Advantages: repair & remanufacture of turbine blades, dies; add features onto existing parts; large builds; functionally graded material; high deposition rate; use of 4-5 axis robots.
- Limitations: lower accuracy & surface finish (needs machining), thermal distortion, limited fine features, no support structures for overhangs (need multi-axis), costly.

Other AM processes (for completeness)

- Material jetting (PolyJet / MJM): inkjet heads deposit droplets of photopolymer / wax that are UV-cured at once; multi-material & full colour, very smooth (16 micron layers); gel-like support dissolved by water jet; parts less durable. Uses: realistic prototypes, medical models, overmoulds.
- Binder jetting (3DP): print head sprays liquid binder on a powder bed; green part is depowdered, then infiltrated / sintered. No heat during print - no residual stress; fast, large; sand cores & moulds, metal parts, full-colour gypsum models. Lower strength / density.
- Sheet lamination (LOM, UAM): sheets are bonded (adhesive or ultrasonic

welding) and the contour is cut by laser / blade; cheap, big parts, little distortion; waste & weak Z bond; paper, plastic, metal foil parts, embedded sensors.

Design for Additive Manufacturing (DfAM)

- Exploit freedom: topology optimisation, lattice / cellular structures, conformal cooling channels, part consolidation, organic shapes (generative design).
- Respect limits: minimum wall thickness (0.4–1 mm), minimum hole size, overhang angle < 45 degrees needs support, avoid large flat horizontal areas (warping), provide escape holes for powder / resin, add fillets, orient critical surfaces upwards, leave machining stock for critical fits.
- Support structures: block, cone, lattice, tree - provide anchor against warping and heat sink (metal), cost time & material; minimise by orientation & self-supporting angles.

Common defects in AM parts

Defect	Cause	Remedy
Warping / curling	Residual thermal stress, shrinkage, poor bed adhesion	Heated bed, brim / raft, supports, preheat, stress relief
Porosity (gas / lack of fusion)	Low energy density, wet powder, trapped gas, bad hatch	Optimise power / speed / hatch, dry powder, HIP
Balling / keyholing (metal)	Too high speed / power; unstable melt pool	Adjust energy density, scan strategy
Cracking / delamination	Thermal gradients, poor layer bonding	Preheat, reduce layer cooling, better parameters
Staircase / poor finish	Layer thickness, orientation	Thinner layers, orientation, post finishing
Dimensional inaccuracy	Shrinkage, stair effects, calibration	Scale compensation, calibration, post-machining

Energy / process parameters - quick formulae

- PBF volumetric energy density $E = P / (v\ h\ t)$ [J/mm^3]; build time = (volume / build rate) + recoating time per layer × layers.
- Number of layers n = part height / layer thickness; FDM extrusion volume rate = road width × layer height × speed.

- SLA cure depth $C_d = D_p \ln(E / E_c)$ (D_p = penetration depth, E_c = critical exposure).

Solved example - FDM build time estimate

Part 60 mm tall printed at 0.2 mm layers -> $n = 300$ layers. If each layer takes 4 min (perimeter + infill), build time = 300 x 4 = 1200 min = 20 hours. Doubling layer height to 0.4 mm halves the layers (~ 10 h) but staircase worsens.

9. Post-Processing of AM Parts

Raw AM parts rarely meet final needs; post-processing improves surface finish, dimensional accuracy, strength and removes supports.

Operation	Purpose / methods	Applicable to
Cleaning & depowdering	Remove loose powder (brushing, air, ultrasonic), resin washing in IPA	PBF, SLA, binder jet
Support removal	Break-away, cutting, wire EDM, soluble dissolving (water / chemical), machining	FDM, SLA, metal PBF
Post-curing	UV / thermal cure to full strength	SLA, DLP, PolyJet
Heat treatment / stress relief	Relieve residual stress (eg. 650-800 C for Ti), age hardening, solutionise	Metal AM
HIP (hot isostatic pressing)	Heat + 100-200 MPa gas closes internal pores -> 100 % density, improved fatigue	Metal AM, critical parts
Machining / grinding / EDM	Tight tolerances, threads, bearing seats, datum faces	Metal & DED parts
Surface finishing	Sanding, bead / sand blasting, tumbling, polishing, electro-polishing, chemical / vapour smoothing (acetone for ABS), barrel finishing, coatings, painting, plating	All
Infiltration	Fill pores with epoxy, wax, bronze (binder jet metal), improving strength	Binder jet, SLS
Sintering / debinding	Remove binder & sinter green part in furnace (shrinks 15-20 %)	Binder jet, metal FFF
Inspection	CMM, 3D scan, CT, dye penetrant, density test	All

10. Applications of AM

Sector	Applications
Aerospace & defence	Lightweight brackets, fuel nozzles (LEAP), turbine blades, ducts, satellite parts, UAV frames, repair of blades; Ti & Ni alloys
Automotive	Prototypes, jigs & fixtures, intake manifolds, lightweight custom parts, motorsport components, tooling with conformal cooling
Medical & dental	Patient-specific implants (hip, cranial, spine), surgical guides, prosthetics, hearing aids, dental crowns & aligners, anatomical models for surgical planning, bioprinting of tissues
Tooling & manufacturing	Injection-mould inserts, die-casting cores, sand moulds / cores (binder jet), investment patterns, end-of-arm grippers, rapid tooling
Consumer & art	Jewellery, eyewear, footwear midsoles, fashion, sculptures, toys, customised products
Construction & others	Concrete 3D printed houses, bridges; food printing; education; energy (heat exchangers); electronics (printed circuits); spare parts on demand

Trends & emerging areas

- Bioprinting: living cells + hydrogel bio-ink to print tissue scaffolds, skin, cartilage.
- 4D printing: printed structure that changes shape / property with time under stimulus (heat, water, light).
- Large-format & construction printing, hybrid AM-CNC machines, in-situ monitoring with AI, multi-laser machines, sustainable (recycled powder), on-demand spare parts, space manufacturing.
- Standards: ASTM F42, ISO / ASTM 52900 (terminology); qualification of aerospace and medical parts is key.

Quick viva / 2-mark questions

- What is STL? -> Standard Tessellation Language: surface model approximated with triangles.
- Name the 7 ASTM categories. -> VPP, ME, PBF, DED, MJ, BJ, SL.
- Why supports? -> To hold overhangs, anchor part and conduct heat. What is HIP? -> Hot isostatic pressing to remove pores.
- Difference SLS vs SLM? -> Sintering (partial fusion, polymers) vs full melting

(metals, dense).

- Which AM suits repair of a turbine blade? -> DED (laser cladding).

Process comparison - quick revision

- SLA / DLP: liquid resin + UV; best finish & accuracy; supports; brittle.
- FDM: filament extrusion; cheap, strong plastics; layer lines; anisotropy.
- SLS / SLM / EBM: powder + laser / e-beam; dense functional parts; metals; supports (metal); costly.
- DED: wire / powder + focused energy; large parts, repair; poor finish.
- Trade-off always: speed vs accuracy vs cost vs material range.

Important exam questions - Unit V

- Explain the steps involved in additive manufacturing with a neat chain diagram. (10 M)
- Classify AM processes (ASTM F42). Advantages & limitations of AM. Materials used. (10 M)
- VAT photo-polymerization (SLA / DLP) - process, parameters, merits & demerits. (10 M)
- FDM / extrusion-based AM with sketch; applications. (10 M)
- Powder bed fusion (SLS, SLM, EBM); DED processes. (10 M each)
- Post-processing of AM parts; applications of AM in different fields. (5-10 M)